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Macroeconomic Information Acquisition Around Earnings Clusters

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ABSTRACT

Research shows that investors acquire and process less firm-specific information on days when many firms announce earnings (hereafter earnings clusters). We show that investors gather more *macroeconomic* information during earnings clusters, that this behavior is amplified during negative economic shocks and concurrent macroeconomic announcements, and that these information acquisition patterns have implications for equity valuations. Our

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findings are consistent with the benefit of extracting macroeconomic information from earnings announcements increasing during earnings clusters, which we confirm empirically. Thus, our results help explain why investors focus less on individual firm news during earnings clusters: They rationally redirect their attention to the most beneficial information.

JEL codes: D81, D83, E44, G11, G14, M41

Keywords: attention allocation; earnings announcements; information acquisition; rational inattention; macroeconomic information

1. Introduction

Research shows that investors acquire and process less firm-specific information on days when many firms announce earnings (hereafter earnings clusters; Hirshleifer, Lim, and Teoh 2009, Drake, Roulstone, and Thornock 2012, DeHaan, Shevlin, and Thornock 2015). This evidence is consistent with earnings clusters straining investors' limited processing capacity and has been interpreted as investors becoming distracted when earnings occur in clusters. (See Blankespoor, DeHaan, and Marinovic 2020 for a review of this literature.) However, investors' information acquisition need not be restricted to firm-specific (micro) information. The aim of this paper is to study investors' acquisition of macroeconomic information during earnings clusters.

We argue that, during earnings clusters, earnings are more informative about the macroeconomy, and, as a consequence, investors direct their limited attention to acquiring and processing macro information. There is compelling evidence on the interconnected nature of firms' economics (Madsen 2017, Brochet, Kolev, and Lerman 2018, Hann, Kim, and Zheng 2019), and a substantial body of research shows that firms' earnings yield insights into macroeconomic factors, including GDP growth (Konchitchki and Patatoukas 2014), inflation (Shivakumar and Urcan 2017), labor market conditions (Hann, Li, and Ogneva 2021), discount rates (Cready and Gurun 2010), and recessions (Lind 2019). We expect the informativeness of corporate earnings about macro outcomes to rise during earnings clusters because when more firms announce earnings, the firm-specific component in earnings is diversified away, making them a less noisy signal about the aggregate component (Savor and Wilson 2016).

Many earnings announcements on the same day strain investors' attention. Thus, rational but attention-constrained investors may try to create economies of scale in their information acquisition.[1] Because macroeconomic shocks affect every firm in a portfolio (albeit to different extents) and corporate earnings become more informative about the macroeconomy during earnings clusters, attending to macro information

[1] We assume that investors choose whether to focus on firm or macro information by evaluating the costs and benefits of their options (Maćkowiak, Matějka, and Wiederholt 2023).

allows investors to resolve more uncertainty per unit of effort than does delving into firm-specific information (Peng and Xiong 2006, Kacperczyk, Van Nieuwerburgh, and Veldkamp 2016). Based on this line of argument, we expect investors to shift their attention to macroeconomic factors and therefore to process more macro information during earnings clusters. This, in turn, can explain why we observe less processing of individual or firm-specific information by investors during earnings clusters.

We start our empirical investigation by studying whether the informativeness of corporate earnings about macroeconomic outcomes differs between earnings announced during cluster versus noncluster days. We expect this to be the case because when many firms announce earnings contemporaneously, the firm-specific component of earnings is diversified away, and therefore aggregate earnings become a less noisy signal about the macroeconomy (Savor and Wilson 2016). To investigate this prediction, we test whether the aggregate earnings of firms announcing during earnings clusters in a given quarter forecasts macroeconomic conditions (GDP growth or CPI rate) better than the aggregate earnings of firms announcing outside earnings clusters during the same quarter. We find that aggregate return on assets (and return on equity) of firms that announce earnings on cluster days better predict the next quarter's GDP growth than aggregate measures from firms that announce on other days. Thus, the *same* level of aggregate corporate performance (e.g., an aggregate return on assets of 5%) is more informative about the macroeconomy when it results from many earnings announcements, consistent with our prediction.

Next, we examine whether the increased informativeness of corporate earnings about the macroeconomy during earnings clusters induces attention-constrained investors to acquire more macro information. To investigate this possibility, we assemble a database of macro-related Internet search intensity from Google Trends and use this database to study the intensity of macro-related Google searches during earnings clusters. We find that macro-related Internet search does intensify during earnings clusters, and this finding extends to multiple categories of macro-related words, suggesting that investors are gathering more macro information on these days. We also find that Google search intensity for nonmacro words declines during earnings clusters, indicating that investors direct some of their attention away from leisure activities and toward investment-related activities when their attention is strained.

We next probe whether macroeconomic conditions (good versus bad economic states) and events (important macroeconomic announcements) amplify the relation between investors' macro information acquisition and earnings clusters. We expect this relation to strengthen during bad economic states, when the benefit of paying attention to macro information relative to micro information increases for investors (Kacperczyk, Van Nieuwerburgh, and Veldkamp 2016). This is because negative economic conditions are associated with increased macroeconomic uncertainty (e.g., Bloom 2014), which makes it more valuable for investors to acquire information, and a higher price of risk (e.g., Harvey 1989), which makes it costlier for investors to bear risk. We test this prediction by expanding our baseline analysis with an interaction term between earnings clusters and an indicator variable that identifies bad economic states based on the Aruoba-Diebold-Scotti Business Condition Index hosted by the Federal Bank of Philadelphia (Aruoba, Diebold, and Scotti 2009). We find that the intensity of macro-related Internet searches increases during earnings clusters, as in our baseline analysis, and that this relation is one to three times as large during bad economic states, consistent with our expectations.

We also expect the relation between investors' macro information acquisition and earnings clusters to strengthen on days with important macro announcements. These announcements are important conduits of macro information (Savor and Wilson 2013) and can increase the marginal benefit or reduce the marginal cost of processing macro information in earnings for investors. For example, Abdalla and Ogneva 2024 document the existence of attention spillovers from Federal Reserve Bank communications to the monetary-policy information in earnings. We test our prediction by expanding our baseline analysis with an interaction term between earnings clusters and an indicator variable that identifies days with important macro announcements. We continue to find that the intensity of macro-related Internet searches increases during earnings clusters but also observe that this relation is one to two times larger on days with important macro announcements. Jointly considered, these two results further support our economic interpretation of the main findings and provide context for understanding how macro information acquisition changes during earnings clusters.

Our finding that earnings clusters prompt greater acquisition of macro information has implications for the processing of firm information on those days at both the extensive margin (which firms investors pay attention to) and intensive margin (conditional on paying attention to a firm, what information investors collect about the firm's performance). At the extensive margin, research shows that investors acquire and process less firm-specific information during earnings clusters (e.g., DeHaan, Shevlin, and Thornock 2015), particularly for less visible firms (Frederickson and Zolotoy 2016). We argue that this reduction of firm-specific information processing is attenuated for firms whose prospects most strongly correlate with macroeconomic factors, which we call bellwether firms. We argue that investors can more easily interpret the earnings of bellwethers due to the increased macro information they collect during earnings clusters. This is because the higher macro informativeness of aggregate earnings allows investors to better understand the macroeconomy, which generates economies of scale in processing the earnings news of bellwethers (which are more affected by the macroeconomy). Therefore, we expect that investors do not slow down their firm-level information gathering for bellwethers as much as they do for other announcing firms during earnings clusters. Our evidence is consistent with this prediction: while the number of periodic filings (10-Ks and 10-Qs) downloaded from EDGAR for nonbellwethers declines during earnings clusters, this decline is attenuated for announcing bellwethers.[2] The evidence is consistent with investors prioritizing the acquisition of firm-specific information for bellwethers during earnings clusters, possibly because of the aforementioned economies of scale they enjoy from already gathering information on the macroeconomy. At the intensive margin, we argue that investors' commitment to gathering macroeconomic information implies that they are relatively more likely to ask macro-related questions in the conference calls hosted by bellwethers during earnings clusters. Consistent with this prediction, we document that the percentage of macroeconomic words included in the conference calls of bellwethers increases significantly relative to nonbellwethers' conference calls during earnings clusters, and this increase is driven by the question and answer (Q&A) segment of the call. Importantly, this finding does not reflect a general effect, as investors do not ask for more macro information in the conference calls of bellwethers outside of earnings clusters.

So far, our analyses support the prediction that investors direct their attention toward macroeconomic information during earnings clusters, and show ramifications for the processing of firm-specific information. In our final test, we examine the equity valuation implications of our findings by studying whether the two-day raw or abnormal stock market response to earnings news varies with earnings clusters differentially for bellwethers and nonbellwethers. Using a sample of firm earnings announcements, we confirm that the positive stock market response to earnings surprises is attenuated during earnings clusters, as found by Hirshleifer, Lim, and Teoh 2009. We then document that this effect of earnings clusters is muted for bellwethers, consistent with investors paying more attention to them. We also consider whether these equity valuation consequences strengthen when investors are already collecting more information on the macroeconomy, which allows us to gauge whether they attain economies of scale in their information acquisition during earnings clusters. If it is easier for investors to interpret bellwether earnings news during earnings clusters, as we predict, then the market response to earnings surprises for bellwethers during earnings clusters should increase with investors' acquisition of macroeconomic information. This intuition resembles that of Hirshleifer and Sheng 2022-09, who emphasize that macro announcements and corporate earnings can complement each other and that investors may rely more on corporate earnings to gain macro insights on macro announcement days. We repeat our equity valuation analysis separately on days with above- and below-median macro-related Google search intensity. We find that the equity market response to earnings surprises of bellwethers during earnings clusters strengthen when investors are collecting more macro information versus when they are not, consistent with our argument that investors

[2] Our definition of bellwethers is based on the correlation of firms' earnings with GDP, and we control for firm size, both standalone and interacted with earnings clusters. Thus, we are not capturing a mechanical effect of investor attention to large firms.

achieve economies of scale in their information acquisition during earnings clusters.

Our paper shows that investors acquire more macro information during earnings clusters, that this behavior is amplified during negative economic shocks and concurrent macro announcements, and that this information acquisition has important implications for equity valuations. Through these findings, we speak to at least three strands of research. First, as discussed earlier, our paper contributes to the rich literature on the determinants of investors' reactions to earnings announcements. Studies show that behavioral factors lead to less investor attention to firm earnings announcements when those earnings are announced on Fridays (DellaVigna and Pollet 2009) as well as concurrently with low market returns (Gulen and Hwang 2012) or distracting events (Hirshleifer, Lim, and Teoh 2009). Our work differs by showing that investors rationally collect more macro information when earnings news better reflects the economy and they face attention constraints. Therefore, as opposed to prior interpretations that the effect of earnings clusters on investors represents an irrationality, we interpret it as a rational outcome (Pedersen 2015), whereby investors choose to disregard some valuable information (firm-specific) to pay attention to a more valuable type of information (macro; Sims 2003, Maćkowiak, Matějka, and Wiederholt 2023). In so doing, the paper supports the notion advanced by Blankespoor, DeHaan, and Marinovic 2020 that constrained investor rationality likely explains the lower processing of firm information by investors during earnings clusters.

Second, the financial economics literature has long sought to connect the macroeconomy to asset pricing (e.g., Fama 1981). Research in this area finds that macro announcements, such as Federal Open Market Committee (FOMC) decisions and GDP, inflation, and unemployment news, affect investors' acquisition of macroeconomic information (Fisher, Martineau, and Sheng 2022, Abdalla and Ogneva 2024) as well as asset prices (Savor and Wilson 2013,,2014], Ai and Bansal 2018, Pan, Sul, and Wang 2024), and that these effects are magnified by macroeconomic uncertainty (Andrei, Friedman, and Ozel 2023). More directly relevant for our paper, research shows that macro announcements can trigger investors' acquisition of idiosyncratic information (Hirshleifer and Sheng 2022-09), particularly for firms with high institutional ownership (Liu, Peng, and Tang 2023-05). We document that the opposite dynamic holds as well: The contemporaneous release of information by multiple firms stimulates investors' acquisition of macro information.

Third, recent work shows that earnings announcements affect the risk premium (Savor and Wilson 2016, Ben-Rephael et al. 2021) and the risk-return relation (Gilbert, Hrdlicka, and Kamara 2018, Chan and Marsh 2022) in ways consistent with the assumption that investors use earnings news to update their priors not only about the performance of the firm but also about the economy. Our evidence supports this important assumption underlying recent influential research. In this way, we add to the line of work showing that firms' disclosures yield macroeconomic insights (e.g., Nallareddy and Ogneva 2017, Ball, Gallo, and Ghysels 2019, Abdalla and Carabias 2022).

2. *Data and Measurement*

We assert that investors gather more macro information during earnings clusters and that this information acquisition has implications for the processing of firm information at both the extensive margin (which firms investors pay attention to) and intensive margin (conditional on paying attention to a firm, what information investors collect about the firm's performance). Accordingly, we need proxies for investors' macro information acquisition, earnings clusters, and firms' macroeconomic exposure.

2.1 MEASUREMENT OF KEY VARIABLES

2.1.1. Investor Macroeconomic Information Acquisition. We rely on multiple complementary proxies of macro information acquisition. For our main analyses, we follow prior research (Drake, Roulstone, and Thornock 2012, DeHaan, Shevlin, and Thornock 2015) and measure information acquisition with Internet search intensity. We download the daily Google search volume index (SVI) for our dictionary of macro-related words (listed in appendix B). We then average the search intensity measure over all macro words in our dictionary to create a variable that captures a daily time series of investors' aggregate interest in macro information (*MGSVI*). This measure has limitations. For example, it is available only in the time-series, and there is no guarantee that macro word searches are performed by investors or for investment purposes.[3] These limitations notwithstanding, the measure allows us to succinctly proxy for investors' macro information acquisition from multiple sources, something that the literature has struggled to do (e.g., Fisher, Martineau, and Sheng 2022).

We complement the main analyses with tests of investor information acquisition of firms' disclosures. We measure investor acquisition of firms' disclosures using daily EDGAR download intensity (*ESV*), which has been shown to capture investor attention to recent as well as historical filings (Drake, Roulstone, and Thornock 2015). This measure is considered superior to alternatives, such as Google searches for firms' tickers, because it is less prone to measurement error. This is because (1) Google search data rely on the accurate selection of tickers, which may change within a firm over time; (2) there is ambiguity inherent in ticker symbols themselves; and (3) EDGAR unquestionably captures interest in firms' accounting information as opposed to some other motive (i.e., speculation, products,

[3] However, our macro search intensity measure (*MGSVI*) is free from some of the limitations of ticker-based firm GSVI, such as ambiguity of meaning and changes in ticker-firm matching over time.

etc.).[4] Furthermore, EDGAR downloads are more likely to reflect the information acquisition of institutional investors (Drake et al. 2020), who are more sophisticated users of accounting information (Blankespoor, DeHaan, and Marinovic 2020) and therefore more likely to choose rationally which information to acquire during earnings clusters, as per our theoretical framework. Nevertheless, we do provide robustness results for these tests using Google search (*GFSVI*) and do so using the finance-investor subcategory classification of Google search, as recommended by deHaan, Lawrence, and Litjens 2025.

EDGAR download intensity allows us to study investor information acquisition about firms more exposed to aggregate shocks, which can be loosely interpreted as the extensive margin of investors' macro information acquisition (namely, whether they acquire macro information). We then introduce a proxy to capture the intensive margin of macro information acquisition (namely, how much macro information investors acquire). To do so, we download earnings conference calls transcripts from Capital IQ. We then use a bag-of-words approach to calculate the percentage of macro-related words (listed in appendix B) in earnings conference calls (*CC %*). We infer that a larger percentage of macro words indicates increased investor attention to the macro implications of earnings announcements. We report examples of conference calls including macro discussions in appendix C. As the examples show, conference call participants ask managers about their assessment of macroeconomic trends, such as GDP growth, inflation, the effect of inflation on aggregate demand, and macroeconomic uncertainty. Finally, we measure investor acquisition of macro information indirectly through asset prices (Ai et al. 2023). Investor information acquisition should guide investor trading and ultimately affect equity valuations. Based on this idea, we study how the incorporation of earnings news into prices varies with earnings clusters differentially for bellwethers and nonbellwethers and whether investors' macro information acquisition mediates these associations. We postpone a detailed discussion of this analysis to section 4.3.

2.1.2. Earnings Clusters. Following prior research, we measure earnings clusters using the number of concurrent daily earnings announcements (Hirshleifer, Lim, and Teoh 2009, DeHaan, Shevlin, and Thornock 2015). Specifically, we use the decile rank of daily earnings announcement counts within each year-quarter (*EC #EAs*). To construct this proxy, we rely on announcement dates from the intersection of Compustat and I/B/E/S, keeping only the dates for which the two databases agree (Blankespoor, DeHaan, and Marinovic 2020), a procedure that ensures announcement dates are precisely measured at the cost of losing some announcements.

[4] See deHaan, Lawrence, and Litjens 2025 for discussion of the measurement error and biases of Google search measures.

We supplement this proxy with a second measure defined as the decile rank of the value-weighted average of daily earnings announcement counts within each year-quarter, where announcements are weighted using the aggregate size of announcing firms on that day relative to the total size of firms announcing earnings during the quarter (*EC Size EAs*). We use this additional proxy because it incorporates the size of announcers and therefore considers the economic importance and complexity of the announcements, which partially drive investors' information acquisition (Frederickson and Zolotoy 2016, Driskill, Kirk, and Tucker 2020) and capital market outcomes (Gilbert, Hrdlicka, and Kamara 2018).

We re-scale *EC #EAs* and *EC Size EAs* to fall within the interval (0,1), which allows us to interpret the coefficient as the effect of moving from the bottom to the top decile of these variables.

2.1.3. Bellwether Firms. As discussed above, because investors' macro information acquisition is unobservable, we complement our analysis of macro information acquisition with an analysis of firm information acquisition, conditional on firms' exposure to the macroeconomy. To do so, we follow the literature (e.g., Anilowski, Feng, and Skinner 2007, Bonsall, Bozanic, and Fischer 2013) and identify firms more exposed to the macroeconomy as bellwethers. Specifically, we take the correlation between a firm's earnings and detrended real GDP (Baxter and King 1999, Brockman, Liebenberg, and Schutte 2010) from the five quarters prior to a given announcement date and define bellwethers as firms for which this correlation exceeds the 90th percentile.

The advantage to this method is that we rely only on information available to investors on a given earnings announcement date and identify firms with the highest actual correlation with the macroeconomy, as opposed to more ad hoc methods of identifying bellwethers using indirect macro-importance measures, like firm size. We also assess the robustness of our findings to measuring bellwethers with an alternative definition based on the correlation between detrended GDP and firms' sales, as discussed in section 4.1.

2.2 SAMPLE FORMATION AND DESCRIPTIVE STATISTICS

Our analyses rely on two main databases: a time-series database at the daily level and a panel database at the firm-quarter level.

For our primary tests, we build a time-series database at the daily level for Internet searches for macro terms, which is available for the years 2004 to 2019 and includes 3,688 observations. Table 1 shows that, in this sample, *MGSVI,* the Google search index for macro words, is on average 41.93 with a standard deviation of 10.53. We also observe that, on average, there are 64 earnings announcements per day, with a standard deviation of 86. We further observe that the aggregate (quarterly) ROA of announcing firms is 0.01% (standard deviation of 0.02), the average market return is 0.03%

TABLE 1
Summary Statistics

	N	Mean	SD	25%	50%	75%
Quarterly Time-Series Variables						
NGDP Growth$_t$	146	0.050	0.025	0.038	0.050	0.065
CPI Growth$_t$	146	0.007	0.006	0.004	0.007	0.010
Agg ROA Cluster$_t$	146	0.006	0.002	0.004	0.006	0.007
Agg ROA Non-Cluster$_t$	146	0.005	0.003	0.004	0.005	0.007
Agg ROE Cluster$_t$	146	0.030	0.009	0.027	0.032	0.036
Agg ROE Non-Cluster$_t$	146	0.030	0.016	0.025	0.031	0.038
#EAs$_t$	146	245.562	134.868	112.000	253.500	365.000
Daily Time-Series Variables						
MGSVI$_t$	3,688	41.932	10.528	34.531	42.381	49.625
#EAs$_t$	3,688	64.199	86.157	8.000	25.000	86.000
EC #EAs$_t$	3,688	0.492	0.321	0.222	0.444	0.778
EC Size EAs$_t$	3,688	0.499	0.316	0.222	0.444	0.778
Agg ROAt	3,688	0.007	0.017	0.003	0.006	0.012
Market Ret %$_t$	3,688	0.000	0.012	−0.004	0.001	0.005
Macro Announce$_t$	3,688	0.209	0.406	0.000	0.000	0.000
Firm-Quarter Panel Variables						
GFSVI$_{i,t}$	24,534	0.005	1.001	−0.514	−0.181	0.363
ESV$_{i,t}$	100,485	537.020	734.960	50.000	235.000	737.000
ESV Human$_{i,t}$	100,485	75.508	118.748	15.000	37.000	83.000
ESV Robot$_{i,t}$	100,485	455.662	652.918	24.000	176.000	632.000
EC #EAs$_t$	100,485	187.987	115.525	93.000	178.000	274.000
EC Size EAs$_t$	100,485	0.042	0.031	0.016	0.038	0.061
Bell$_{i,t}$	100,485	0.115	0.319	0.000	0.000	0.000
$\lvert Ret \rvert_{i,t}$	100,485	0.032	0.037	0.008	0.020	0.042
B/M$_{i,t}$	100,485	0.491	0.370	0.239	0.414	0.669
Leverage$_{i,t}$	100,485	0.207	0.203	0.017	0.164	0.329
Size$_{i,t}$	100,485	7.579	1.787	6.278	7.519	8.760
#Analysts$_{i,t}$	100,485	9.911	6.944	5.000	8.000	14.000
Macro Announce$_t$	100,485	0.205	0.404	0.000	0.000	0.000
Quarter 4$_{i,t}$	100,485	0.250	0.433	0.000	0.000	0.000
Spread$_{i,t}$	100,485	−0.002	0.003	−0.002	−0.001	−0.000
Volume$_{i,t}$	100,485	20.541	23.950	6.712	12.472	24.152
News Stories$_{i,t}$	100,485	15.027	11.402	8.000	12.000	17.000
Inst Own$_{i,t}$	100,485	0.732	0.274	0.599	0.796	0.925

See appendix A for detailed definitions and sources for all variables used in our tests.

(standard deviation of 1.15), and approximately 21% of days in our sample feature a macroeconomic announcement.

For our complementary analyses, we study patterns in investors' information acquisition using a sample of firms' quarterly earnings announcements that includes 100,485 observations for 4,912 firms over the years 2002 to 2017. We construct this sample by combining multiple databases. We start from a sample of firm-quarter observations from Compustat, which allows us to observe firm characteristics and earnings announcement dates. We merge this database with investor information acquisition from EDGAR, available to us between 2002 and 2017, to measure investor information acquisition of firm SEC filings on their earnings announcement date. We merge the resulting firm-earnings announcement database with I/B/E/S (for analyst following), Ravenpack (for news coverage), and a set of dates corresponding to macro announcements, including news about manufacturing, personal consumption, monetary policy, GDP, unemployment, and inflation. Finally, we eliminate all firms lacking necessary data as well as those having less than one million dollars of assets, stock prices smaller than one dollar, or a lag between fiscal quarter-end and earnings announcement larger than 90 days.

Table 1 reports descriptive statistics for this sample and shows that the average number of daily earnings announcements is 187.99 with a standard deviation of 115.53, similar to related studies (e.g., DeHaan, Shevlin, and Thornock 2015).[5] Moving to our outcome variables, the table shows that, on average, a firm's periodic filings are downloaded from EDGAR 537.02 times on the firm's earnings announcement date, with a standard deviation of 734.96. We also observe for a subsample of 53,247 firm-earnings announcement observations with earnings conference call transcripts that, on average, 2.32% of the words included in those calls relate to the macroeconomy, with a standard deviation of 4.24 percentage points. A little more than half of the conference calls do not include any macro words. Firm characteristics resemble those in related studies. For example, our sample firms have average book-to-market (*B/M*) of 0.49, leverage (*Leverage*) of 0.21, and size (*Size*) of 7.58, similar to the values reported by DeHaan, Shevlin, and Thornock 2015 (0.57, 0.20, and 6.66, respectively). Untabulated descriptive statistics show that bellwethers, which represent approximately 10% of our sample firms, resemble nonbellwethers in terms of firm characteristics. The two firm types have almost identical institutional ownership (0.72 versus 0.71), but bellwethers are smaller (7.31 versus 7.52) and have slightly lower book-to-market (0.42 versus 0.48) and leverage (0.16 versus 0.21).

Finally, we construct a time-series database at the quarterly level to test the macroeconomic content of earnings announced during earnings cluster days and nonearnings cluster days. We postpone a discussion of how this database is constructed to section 3.1.

3. Macro Information Acquisition During Earnings Clusters

We expect that, during earnings clusters, earnings are more informative about the macroeconomy, and, as a consequence, investors direct more of their limited attention to acquiring and processing macro information. We also expect these information acquisition patterns on earnings cluster days

[5] The difference in average daily earnings announcements between this sample and the daily time series exists because the time-series sample includes all days, even those with zero announcements, while the panel data, by construction, contain no zeros because every observation is an earnings announcement. Further, in the panel data every day with n earnings announcements corresponds to n observations, one for each firm announcing on that day.

to be amplified during bad economic states and when important macroeconomic data are concurrently announced.

3.1 THE MACROECONOMIC CONTENT OF EARNINGS DURING EARNINGS CLUSTERS

Our prediction that investors acquire more macro information during earnings clusters assumes that the informativeness of corporate earnings about macroeconomic outcomes is larger during earnings clusters. We assume this for two reasons. First, corporate earnings have both a firm-specific and a macro component. When many firms announce earnings contemporaneously, the firm-specific component is diversified away, and therefore aggregate earnings become a less noisy signal about the macroeconomy (Savor and Wilson 2016, Lind 2019). Thus, we might expect that aggregate earnings announced during clusters contain more macro information by virtue of the fact that more firms announce earnings on these days. Second, from a Bayesian perspective, information is defined in terms of its effect on priors. If, in response to many firms announcing earnings, investors update their priors about the macroeconomy more (i.e., more contemporaneous announcements trigger a larger update than fewer announcements), then more macro information enters the market on those days (e.g., Mankiw and Reis 2002). Either story predicts an increase in the macro content of earnings during earnings clusters.

We test this prediction empirically using the following regression:

$$\text{Macro Growth}_t = \alpha + \beta_1 \text{ Earn Cluster}_t + \beta_2 \text{ Earn Non-Cluster}_t + \beta_3 \# \text{ EAs}_t + \eta_q + \epsilon_t. \quad (1)$$

We form a quarterly time-series sample of two macroeconomic outcomes: growth in nominal gross domestic product (*NGDP Growth*) and growth in the consumer price index (*CPI Growth*). The sample spans the years 1982 to 2019 and includes 146 quarters. Descriptive statistics reported in table 1 indicate that, during our sample period, growth in NGDP (CPI) is on average 5.0% (0.7%), with a standard deviation of 2.5 percentage points (0.6 percentage points). We add to this database measures of earnings announced during earnings clusters (*Earn Cluster*) and earnings announced outside earnings clusters (*Earn Non-Cluster*) during the quarter. We measure earnings announced during a cluster (outside a cluster) using the quarterly aggregate scaled earnings—ROA or ROE—of firms that announced on days with a number of announcements above (below) the quarterly median. Interestingly, earnings announced during and outside earnings clusters are quite similar: The mean aggregate ROA (ROE) announced during earnings clusters is 0.6% (3.0%) compared to 0.5% (3.0%) mean aggregate ROA (ROE) announced outside earnings clusters. Thus, there do not appear to be meaningful systematic differences in performance in the two types of days.

We perform our analyses by regressing the two macroeconomic outcomes on earnings announced during and outside earnings clusters. Importantly, we also control for the number of firms announcing earnings on any given day. Using announcers' aggregate ROA or ROE and controlling for the number of earnings announcements allow us to compare the information content of a one-unit increase in earnings for cluster versus noncluster days, irrespective of the number of announcing firms. Thus, any differences in the coefficients for earnings announced on cluster versus noncluster days can be attributed to the differing macro information content of those earnings and not to a mechanical dollar increase in earnings due to the presence of more announcements.

We determine the autocorrelation structure to estimate equation (1) using the Bayesian information criterion with a preference for the most parsimonious model, although results are robust to permutations of this choice. We then employ ARMAX estimation to correct parameter estimates and standard errors for autocorrelation. We also check for stationarity of all time-series variables using the Dickey-Fuller and Phillips-Perron unit root tests before estimation. We find that these tests strongly reject the null hypothesis of the presence of a unit root for all variables, which reduces concerns about spurious correlation driving our results. For ease of interpretation, we standardize all continuous variables to have a mean of zero and a standard deviation of one.

We present results from these regressions in table 2. Columns 1 and 2 (3 and 4) report results for regressions with *NGDP Growth* (*CPI Growth*) as the dependent variable. The results indicate that, in every instance, aggregate earnings from firms announcing during clusters in the quarter better predict growth in both NGDP and CPI than the aggregate earnings from firms announcing during nonclusters. The estimated β_1 coefficient is always positive, statistically significant, and statistically larger than the estimated β_2 coefficient at the 5% or 10% levels. (The p-value for the difference in coefficients is indicated in the table.) These results are consistent with an increase in the macro content of earnings during earnings clusters, on which our prediction that investors acquire more macro information during earnings clusters is predicated.

The evidence above suggests that earnings announced during earnings clusters contain more information about the macroeconomy, making it more beneficial for investors to pay attention to macro information on those days. Absent capacity constraints, investors could simply expand their macro information acquisition without necessarily reducing their firm-specific information acquisition. However, research shows that investors reduce their acquisition of firm-specific information during earnings clusters, consistent with the presence of attention capacity constraints. Along these lines, we observe in table 1 of the online appendix that, during earnings clusters, market-level uncertainty, as measured with the VIX index, declines (panel A) while firm-specific uncertainty, as measured with idiosyncratic volatility, increases (panel B). Overall we interpret these findings as suggesting that earnings clusters make it more valuable for investors to acquire and process macro information.

TABLE 2

The Macro Information in Earnings Clusters Versus Non-Clusters

$\text{Macro Growth}_t = \alpha + \beta_1 \text{ Earn Cluster}_t + \beta_2 \text{ Earn Non-Cluster}_t + \beta_3 \text{ EAs}_t + \eta_q + \varepsilon_t$

Dep Var =	*NGDP Growth$_t$*		*CPI Growth$_t$*	
	(1)	(2)	(3)	(4)
Agg ROA Cluster$_t$	4.951***		0.472**	
	(3.52)		(1.98)	
Agg ROA Non-Cluster$_t$	0.999*		0.064	
	(1.96)		(0.44)	
Agg ROE Cluster$_t$		0.923***		0.113*
		(3.66)		(1.92)
Agg ROE Non-Cluster$_t$		0.143		0.005
		(1.47)		(0.19)
#EAs$_t$	−0.000	−0.000***	−0.000***	−0.000***
	(−0.30)	(−2.59)	(−2.88)	(−4.50)
p-value of $\beta_1 = \beta_2$:	0.01	0.01	0.09	0.08
AR Terms Included:	1	1	2	2
Quarter Fixed Effects:	Yes	Yes	Yes	Yes
Pseudo R^2	0.33	0.33	0.41	0.41
N Quarters	146	146	146	146

Estimation: ARMAX estimation. Robust standard errors are implicitly adjusted for autocorrelation through ARMAX. Statistical significance is denoted as follows: $^*p < 0.10, **p < 0.05, ***p < 0.01$ (two-tailed), t statistics are in parentheses. p-values for $\beta_1 = \beta_2$ are based on one-tailed tests. **Dependent Variable**: is either the final vintage of quarterly NGDP growth (*NGDP Growth$_t$*) or the growth rate in the total CPI index (*CPI Growth$_t$*) for quarter t. **Variables of Interest**: include aggregate ROA (ROE) from earnings clusters (*Agg ROA (ROE) Cluster$_t$*), and from non-clusters (*Agg ROA (ROE) Non-Cluster$_t$*) in quarter t. **Control Variables**: include the number of firms announcing earnings in quarter t (*#EAs$_t$*).

3.2 MACRO-ORIENTED INTERNET SEARCHES DURING EARNINGS CLUSTERS

Next, we test whether investors acquire more macro information on earnings cluster days. As described in section 2.1.1, we use the Internet search intensity of macro-related words, which is available at the daily frequency, as the dependent variable in time-series regressions of the following form:

$$\text{Y}_t = \alpha + \beta \text{ EC}_t + \sum \gamma_j \text{ Controls}_{j,t} + \rho_d + \epsilon_t. \tag{2}$$

We rely on our dictionary of macro-oriented words discussed in appendix B and download daily Google search intensity for each term in the dictionary. In addition to forming an overall measure of macroeconomic search intensity based on all the words in our dictionary (*MGSVI*), we also categorize these words to create measures of different macro topics, including aggregate demand (*MGSVI Demand*), aggregate supply (*MGSVI Supply*), aggregate output (*MGSVI Outcomes*), policy (*MGSVI Policy*), and equity markets (*MGSVI Equity*). We also perform the analysis on a category of nonmacro-related words to rule out that our findings reflect an increase in all Internet searches in those days (*Non-Macro GSVI*).

We account for important determinants of macro information acquisition by controlling for the direction of earnings news with the aggregate ROA of announcing firms ($Agg\ ROA_t$), for investors' differential information acquisition preferences on low market return days (Gulen and Hwang 2012) with market returns on the day of the announcement ($Market\ Ret_t$), and for general macro information with an indicator for days on which an important macroeconomic announcement is made ($Macro\ Announce_t$). We also include day-of-week fixed effects to control for recurring attention and trading patterns throughout the week (DellaVigna and Pollet 2009). As in the previous section, (1) we verify the stationarity of all time-series variables using Dickey-Fuller and Phillips-Perron unit root tests and find that these tests strongly reject the null hypothesis of the presence of a unit root for all variables; (2) we employ ARMAX estimation to correct parameter estimates and standard errors for autocorrelation; and (3) we choose lags using the Bayesian information criterion. Finally, for ease of interpretation, we standardize all continuous variables to have a mean of zero and a standard deviation of one.

We expect to observe an increase in macro information acquisition through Internet search during earnings clusters if investors pay attention to the macroeconomy on these days. Thus, we expect to observe a positive β coefficient estimate for our measures of earnings clusters ($EC\ \#EAs_t$ and $EC\ Size\ EAs_t$). Table 3 consistently documents a positive association between the intensity of macro search and earnings clusters, irrespective of whether we use a comprehensive index of macro search terms (columns 1 and 2), categories of macro search terms (columns 3 to 7), or individual search terms (untabulated) as the dependent variable and irrespective of how we measure earnings clusters. Column 1 indicates that moving from the bottom to the top decile in *EC #EAs* is associated with a 0.34 standard deviation increase in *MGSVI* or an 8.6% increase relative to the sample mean.[6] Finally, Table 3 also shows that the search of nonmacro terms (e.g., "pet food," "oil change," "NFL," "MLB") has a negative relationship with concurrent earnings announcements (column 8). This finding indicates that we are not simply capturing days with more Internet searches. Moreover, it also suggests that, during earnings clusters, investors may expand their attention capacity for work-related information acquisition at the expense of nonwork-related information acquisition, as suggested by Andrei, Friedman, and Ozel 2023.

3.3 AMPLIFYING IMPACT OF MACROECONOMIC SHOCKS AND EVENTS

We probe whether macroeconomic shocks and events amplify the relation between investors' macro information acquisition and earnings clusters. Specifically, we exploit two sources of variation in the incentive

[6] The within-day of the week standard deviation in *MGSVI* is almost identical to its raw standard deviation.

TABLE 3

Macro Information Acquisition from Other Sources and Earnings Clusters

	$Y_t = \alpha + \beta\ \mathrm{EC}_t + \sum \gamma_j\ \mathrm{Controls}_{j,t} + \rho_d + \epsilon_t$							
$Y_t =$	*MGSVI$_t$*		*MGSVI Demand$_t$*	*MGSVI Supply$_t$*	*MGSVI Outcome$_t$*	*MGSVI Policy$_t$*	*MGSVI Equity$_t$*	*Non-Macro GSVI$_t$*
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EC #EAs$_t$	0.344***		0.186***	0.264***	0.281***	0.369***	0.116	−0.078***
	(6.07)		(2.91)	(4.40)	(4.80)	(4.90)	(1.51)	(−3.16)
EC Size EAs$_t$		0.149***						
		(3.64)						
Agg ROA$_t$	0.018**	0.015**	0.015*	0.014	0.015*	0.019**	0.016	−0.002
	(2.28)	(1.98)	(1.79)	(1.50)	(1.82)	(2.16)	(1.25)	(−0.47)
Market Ret$_t$	−0.007	−0.009	0.016*	−0.008	−0.014	−0.009	−0.010	0.004
	(−0.99)	(−1.17)	(1.83)	(−0.91)	(−1.54)	(−0.70)	(−0.68)	(1.58)
Macro Announce$_t$	0.050***	0.045***	0.024	−0.027	−0.016	0.220***	−0.026	−0.008
	(2.73)	(2.46)	(0.95)	(−1.07)	(−0.75)	(7.47)	(−0.82)	(−0.97)
AR Terms Included:	7	7	7	7	7	7	7	7
Day-of-Week FEs:	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N Days	3,688	3,688	3,688	3,688	3,688	3,688	3,688	3,688

Estimation: ARMAX estimation with lags of the dependent variable selected using the Bayesian information criterion, with a preference for the most parsimonious model. Robust standard errors are implicitly adjusted for autocorrelation through ARMAX. All continuous variables are standardized. Statistical significance is denoted as follows: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (two-tailed), t statistics are in parentheses. Intercept suppressed. **Dependent Variable**: Y_t is either *MGSVI$_t$*, the daily Internet search intensity for all macro terms in our dictionary, *MGSVI Category$_t$*, the daily Internet search intensity for macro terms in the following categories: *Demand*, *Supply*, *Outcome*, *Policy*, or *Equity*, or *Non-Macro Search$_t$*, the daily Internet search intensity for non-macro terms in our dictionary (see appendix B). **Variable of Interest**: *EC #EAs$_t$*, the decile rank of the number of firms announcing earnings on day t, or *EC Size EAs$_t$*, the decile rank of the size-weighted number of firms announcing earnings on day t. **Control Variables**: *Agg ROA$_t$* is the aggregate ROA of announcing firms; *Market Ret$_t$* is the market return on day t; and *Macro Announce$_t$* is an indicator identifying days in which monetary policy, labor, inflation, or GDP information is released. Details on our measures of Internet search intensity are provided in appendix B.

to acquire macro information: good versus bad states of the economy and macro announcement days versus nonmacro announcement days.

First, we study whether our results vary with the state of the economy (good versus bad). Building on Kacperczyk, Van Nieuwerburgh, and Veldkamp 2016, who show analytically and empirically that mutual fund managers optimally choose to process macro information in recessions (and micro information in booms) because recessions are associated with increased uncertainty (Bloom 2014) and with a higher price of risk (Harvey 1989), we expect our findings to strengthen during negative states of the economy. To test this prediction, we expand equation (2) by including an indicator that identifies negative economic states (*Bad*), both standalone and interacted with our measures of earnings clusters. We identify bad economic states using the Aruoba-Diebold-Scotti Business Condition Index (Aruoba, Diebold, and Scotti 2009). This index, hosted by the Federal Reserve Bank of Philadelphia, is designed to track real business conditions at high observation frequency. Positive values indicate better-than-average conditions, while negative values indicate worse-than-average conditions. We exploit this feature to identify bad economic states as days of large (bigger than two standard deviations) negative revisions in economic conditions.[7]

We report estimated coefficients in table 4, column 1 for *EC #EAs* and column 3 for *EC Size EAs*. We find that investors acquire more macro information during earnings clusters (estimated $\beta_1 > 0$ and statistically significant at the 1% level), consistent with our findings in section 3.2. Importantly, we also observe that the estimated coefficients on the interaction between earnings clusters and bad economic states (β_2) is also positive and statistically significant at the 10% or 5% level, consistent with investors increasing their macro information acquisition during earnings clusters even more in bad economic states. The economic magnitude is meaningful, with Google searches for macro-related terms during bad economic states being roughly one (column 1) to three (column 3) times larger than during good economic states.

Second, we study whether our results vary with important macro announcements. We expect that investors gather even more macro information during earnings clusters that overlap with macro announcements because the announcements not only convey important macro information to the market (Savor and Wilson 2013) but they also affect the benefits and costs of attending to macro information. For example, Abdalla and Ogneva 2024 show that aggregate earnings response coefficients are negative and statistically significant only during the FOMC meeting weeks, when investors associate higher aggregate earnings with future monetary policy tightening and higher risk premiums, consistent with the existence of attention spillovers from Fed communications to the monetary-policy–relevant information in earnings. To test our prediction, we expand equation (2) by

[7] Our results are robust to alternative definitions of bad economic states.

TABLE 4
Macro Information Acquisition by Economic Status/Announcement

$$\text{MGSVI}_t = \alpha + \beta_1\ \text{EC}_t + \beta_2\ \text{EC}_t \times \text{Z}_{i,t} +\ \text{Z}_t + \beta_3 + \sum \gamma_j\ \text{Controls}_{j,t} + \rho_d + \epsilon_t$$

$\text{EC}_t =$	*EC #EAs$_t$*		*EC Size EAs$_t$*	
	(1)	(2)	(3)	(4)
EC_t	0.334***	0.308***	0.131***	0.120***
	(6.01)	(5.46)	(3.32)	(3.02)
$EC_t \times Bad_t$	0.281*		0.361**	
	(1.94)		(2.49)	
$EC_t \times$ Macro Announce$_t$		0.221***		0.263***
		(4.09)		(4.92)
Bad_t	−0.127		−0.165**	
	(−1.57)		(−2.03)	
Macro Announce$_t$	0.059***	−0.047	0.054***	−0.072**
	(3.31)	(−1.43)	(3.00)	(−2.21)
Agg ROA$_t$	0.014*	0.013*	0.012	0.012
	(1.89)	(1.80)	(1.62)	(1.54)
Market Ret$_t$	−0.005	−0.006	−0.006	−0.007
	(−0.75)	(−0.83)	(−0.88)	(−0.97)
AR Terms Included:	7	7	7	7
Day-of-Week FEs:	Yes	Yes	Yes	Yes
N Days	3,688	3,688	3,688	3,688

Estimation: ARMAX estimation with year-week fixed effects. Robust standard errors are implicitly adjusted for autocorrelation through ARMAX. All continuous variables are standardized. Statistical significance is denoted as follows: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (two-tailed), t statistics are in parentheses. Intercept suppressed. **Dependent Variable**: *$MGSVI_t$*, the Google search intensity index for words in our macro dictionary during quarter t. **Variable of Interest**: *EC_t* is either *EC #EAs$_t$*, the decile rank of the number of firms announcing earnings on day t, or *EC Size EAs$_t$*, the decile rank of the size-weighted number of firms announcing earnings on day t (both re-scaled to fall between (0,1)). *Bad_t* is an indicator that identifies negative economic states as days with large declines in the Aruoba-Diebold-Scotti Business Condition Index (Aruoba, Diebold, and Scotti 2009). *Macro Announce$_t$* is an indicator identifying days in which monetary policy, labor, inflation, or GDP information is released. **Control Variables**: *Agg ROA$_t$* is the aggregate ROA of announcing firms; *Market Ret$_t$* is the market return on day t. Details on our measures of Internet search intensity are provided in appendix B.

interacting the variable *Macro Announce$_t$*, described above, which identifies macro announcement days, with our measures of earnings clusters.

We report estimated coefficients in table 4, column 2 for *EC #EAs* and column 4 for *EC Size EAs*. We find that investors acquire more macro information during earnings clusters (estimated $\beta_1 > 0$ and statistically significant at the 1% level), consistent with our findings in section 3.2. Importantly, we also observe that the estimated coefficient on the interaction between earnings clusters and macroeconomic announcement days (β_2) is also positive and statistically significant at the 1% level, consistent with investors increasing their macro information acquisition during earnings clusters that overlap with macro announcements even more than during earnings clusters that do not. The economic magnitude is meaningful, with Google searches for macro-related terms during earnings clusters overlapping with macro announcements being roughly one (column 2) to two (column 4) times larger than during earnings clusters not overlapping with macro announcements.

These two analyses suggest that the relation between investors' macro information acquisition and earnings clusters varies as predicted by economic theory. Accordingly, these analyses reinforce the economic interpretation of our findings.

4. Acquisition of Macro and Firm Information During Earnings Clusters

In this section, we explore the implications of our findings for the processing of firm-specific information at both the extensive and intensive margin as well as potential equity valuation consequences.

4.1 EDGAR DOWNLOAD INTENSITY AROUND EARNINGS ANNOUNCEMENTS DURING EARNINGS CLUSTERS

Research shows that investors acquire and process less firm information during earnings clusters (e.g., DeHaan, Shevlin, and Thornock 2015), particularly for less visible firms (Frederickson and Zolotoy 2016). We argue that this reduction of firm information processing during earnings clusters may be attenuated for what we term bellwether firms. This is because investors can more readily interpret their earnings news in the context of the increased macroeconomic information collected during earnings clusters. Therefore, we expect that investors do not slow down their information gathering for bellwethers as much as they do for other announcers during earnings clusters. We test this prediction by estimating the following regression:

$$\mathrm{EDGAR}_{i,t} = \alpha + \beta_1\ \mathrm{EC}_t \times\ \mathrm{Bell}_{i,t} + \beta_2\ \mathrm{EC}_t + \beta_3\ \mathrm{Bell}_{i,t} + \sum \gamma_j\ \mathrm{Controls}_{j,i,t} + \psi_w + \epsilon_{i,t}. \tag{3}$$

We measure firm information acquisition ($EDGAR_{i,t}$) using the number of EDGAR downloads for a firm's periodic filings (10-Ks and 10-Qs) around its earnings announcement on day t ($ESV_{i,t}$). We also decompose this variable into two components, one for download intensity from IP addresses not suspected of being robots or automated web crawlers ($ESV\ Human_{i,t}$) and one for the download intensity from IPs suspected of being robots ($ESV\ Robot_{i,t}$). We do not make a differential prediction for robot versus nonrobot IP search intensity for two reasons: because automated downloads may be programmed to access EDGAR files conditional on other inputs to the program, such as the number of concurrent earnings announcements, and because they may serve information providers catering to investors (Andrei, Friedman, and Ozel 2023). The two variables of interest are EC_t, our proxy for earnings clusters, and $Bell_{i,t}$, which identifies bellwethers, both measured as described in section 2.1.

The regression includes controls from related prior work (Drake, Roulstone, and Thornock 2012, 2015), deHaan, Lawrence, and Litjens 2025): absolute returns ($|Ret|_{i,t}$), book-to-market ratio ($B/M_{i,t}$), financial leverage ($Leverage_{i,t}$), firm size ($Size_{i,t}$), analyst following ($\#Analysts_{i,t}$), bid-ask spread

($Spread_{i,t}$), trading volume ($Volume_{i,t}$), news coverage ($News\ Stories_{i,t}$), and institutional ownership ($Inst\ Own_{i,t}$) as well as an indicator for a firm's fourth-quarter earnings ($Quarter\ 4_{i,t}$) and an indicator for days with a macro announcement, including monetary policy, labor, inflation, and GDP announcements ($Macro\ Announce_t$). Moreover, we interact each of these controls with EC_t to ensure that any different results for bellwethers are not driven by other observable differences between bellwethers and other firms (e.g., differences in average size). We also absorb year-week fixed effects (ψ_w) because investor information acquisition follows time trends (Drake et al. 2017) that can be reduced only through careful selection of time fixed effects (e.g., deHaan, Lawrence, and Litjens 2025).[8] We do not include firm fixed effects, because the interpretation of results becomes difficult when pairing panel ID fixed effects and high-frequency time fixed effects (Kropko and Kubinec 2020), but show that our inference is robust to including firm fixed effects. Finally, we cluster standard errors at the year-quarter level because one of our variables of interest, EC_t, is constructed at that level, inducing within-year-quarter correlation in the error term.[9]

Two coefficient estimates are of interest to us. The first is β_2, which describes the association between investor information acquisition and earnings clusters for nonbellwethers. We expect the estimate of this coefficient to be negative, consistent with evidence that investors acquire less firm information on busy earnings announcement days (e.g., DeHaan, Shevlin, and Thornock 2015). The second coefficient of interest is β_1, which describes how the relation between investor acquisition of firm information and earnings clusters differs for bellwethers. We expect this coefficient estimate to be positive, suggesting that the predicted negative association between investor information acquisition and earnings clusters is muted for firms with the greatest exposure to aggregate shocks. This finding would be consistent with investors more easily interpreting the earnings of bellwether firms when they contemporaneously increase macroeconomic information collection during earnings clusters.

We present estimated coefficients in table 5. In the first three columns, we measure earnings clusters with $EC\ \#EAs_t$ and $EDGAR_{i,t}$ with $ESV_{i,t}$, $ESV\ Human_{i,t}$, and $ESV\ Robot_{i,t}$, respectively. Column 1 reports a statistically negative β_2 estimate, consistent with investors acquiring less firm information during earnings clusters. The estimated coefficient is sizable and suggests that moving from the bottom to the top decile of $EC\ \#EAs_t$ is associated with a 0.221 standard deviation (0.308 within-fixed effects standard deviation) decline in $ESV_{i,t}$, which corresponds to a 30% decline relative to the unconditional sample mean. Importantly, the column also documents

[8] We absorb fixed effects using the Stata command *reghdfe* from Correia 2023.

[9] We do not cluster by firm as well as by year-quarter because of the evidence of Conley, Goncalves, and Hansen 2018 that two-way clustering can perform poorly. In table 2 of the online appendix, we find that our results are qualitatively similar when clustering by firm, firm and year-quarter, year-week, and firm and year-week.

TABLE 5
Investor Attention to Bellwethers and Earnings Clusters

$$\text{EDGAR}_{i,t} = \alpha + \beta_1\ \text{EC}_t \times\ \text{Bell}_{i,t} + \beta_2\ \text{EC}_t + \beta_3\ \text{Bell}_{i,t} + \sum \gamma_j\ \text{Controls}_{j,i,t} + \psi_w + \epsilon_{i,t}$$

$\text{EC}_t =$	*EC #EAs$_t$*			*EC Size EAs$_t$*		
	(ESV)	(ESV Human)	(ESV Robot)	(ESV)	(ESV Human)	(ESV Robot)
$EC_t \times Bell_{i,t}$	0.101**	0.166***	0.089**	0.077*	0.103***	0.071
	(2.27)	(3.28)	(2.10)	(1.81)	(3.40)	(1.58)
EC_t	−0.222***	−0.175**	−0.195***	−0.009	−0.080*	0.016
	(−3.00)	(−2.09)	(−2.88)	(−0.17)	(−1.69)	(0.29)
$Bell_{i,t}$	−0.111**	−0.149***	−0.104**	−0.086**	−0.089***	−0.084**
	(−2.66)	(−3.23)	(−2.56)	(−2.32)	(−3.13)	(−2.16)
$\|Ret\|_{i,t}$	0.007	0.044**	0.008	0.026	0.058***	0.023
	(0.35)	(2.36)	(0.41)	(1.22)	(3.41)	(1.15)
$B/M_{i,t}$	0.071***	0.041***	0.071***	−0.027***	−0.037***	−0.027**
	(5.38)	(2.79)	(4.75)	(−2.74)	(−2.77)	(−2.41)
$Leverage_{i,t}$	0.093***	0.144***	0.075***	0.054***	0.068***	0.047***
	(3.75)	(4.44)	(3.81)	(3.50)	(4.57)	(3.19)
$Size_{i,t}$	−0.108**	0.007	−0.111***	−0.026	0.124***	−0.047*
	(−2.59)	(0.20)	(−2.87)	(−0.99)	(5.18)	(−1.76)
$\#Analysts_{i,t}$	0.014	0.061***	0.012	−0.030**	0.058***	−0.039***
	(0.77)	(3.19)	(0.62)	(−2.43)	(3.95)	(−2.77)
$Spread_{i,t}$	0.067**	0.018	0.068**	0.023	0.010	0.023
	(2.04)	(0.52)	(2.15)	(1.44)	(0.44)	(1.46)
$Volume_{i,t}$	−0.084***	−0.036*	−0.091***	−0.040**	0.019	−0.049***
	(−4.67)	(−1.82)	(−4.84)	(−2.38)	(1.19)	(−2.90)
$News\ Stories_{i,t}$	0.308***	0.644***	0.226***	0.187***	0.405***	0.130***
	(9.46)	(15.90)	(7.00)	(6.18)	(11.82)	(4.33)
$Inst\ Own_{i,t}$	0.150***	0.042*	0.155***	0.122***	0.018	0.130***
	(4.86)	(1.77)	(5.17)	(5.02)	(1.20)	(5.32)
$Quarter\ 4_{i,t}$	0.047	−0.004	0.053	0.015	−0.018	0.014
	(1.16)	(−0.07)	(1.40)	(0.29)	(−0.42)	(0.28)
$Macro\ Announce_t$	0.008	−0.087*	0.031	0.019	−0.020	0.031
	(0.24)	(−1.74)	(0.90)	(0.57)	(−0.44)	(0.95)
p-value of $\beta_1 + \beta_2 = 0$:	0.06	0.91	0.07	0.14	0.59	0.07
Controls × *EC_t*	Yes	Yes	Yes	Yes	Yes	Yes
Year-Week FEs:	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.55	0.45	0.54	0.55	0.45	0.54
N Firm-Quarters	100,485	100,485	100,485	100,485	100,485	100,485

Estimation: Fixed-effects (within) estimation with year-week fixed effects. Standard errors are clustered by year-quarter. All continuous variables are standardized. Statistical significance is denoted as follows: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (two-tailed), t statistics are in parentheses. Intercept suppressed. **Dependent Variable**: $EDGAR_{i,t}$ is $ESV_{i,t}$, $ESV\ Human_{i,t}$, or $ESV\ Robot_{i,t}$, the EDGAR search intensity for firm i on its earnings announcement day t attributable to all, non-robot, or robot IP addresses, respectively. **Variable of Interest**: EC_t is either *EC #EAs$_t$*, the decile rank of the number of firms announcing earnings on day t, or *EC Size EAs$_t$*, the decile rank of the size-weighted number of firms announcing earnings on day t (both re-scaled to fall between (0,1)). $Bell_{i,t}$ is an indicator for firm i being designated a bellwether. **Control Variables**: Absolute returns ($|Ret|_{i,t}$), book-to-market ratio ($B/M_{i,t}$), financial leverage ($Leverage_{i,t}$), firm size ($Size_{i,t}$), analyst following ($\#Analysts_{i,t}$), bid-ask spread ($Spread_{i,t}$), trading volume ($Volume_{i,t}$), news coverage ($News\ Stories_{i,t}$), institutional ownership ($Inst\ Own_{i,t}$), an indicator for a firm's fourth-quarter EA ($Quarter\ 4_{i,t}$), and an indicator identifying days in which monetary policy, labor, inflation, or GDP information is released ($Macro\ Announce_t$). Each control is also interacted with EC_t.

a statistically positive and economically meaningful β_1 estimate, indicating that the negative association between earnings clusters and investor information acquisition is muted for bellwethers, which have the greatest exposure aggregate shocks. The economic magnitude is large, suggesting that decline in $ESV_{i,t}$ associated with moving from the bottom to the top decile of $EC\ \#EAs_t$ is 46% smaller for bellwethers. Nonetheless, the sum of the two coefficients in column 1 is negative and statistically significant at the 10% level (p-value = 0.06), indicating that investors reduce their firm-specific information acquisition for bellwethers as well. Coefficient estimates for control variables are consistent with economic intuition: Investors download more EDGAR filings on earnings announcement days for riskier and more leveraged firms and for firms with more institutional investors and more media coverage.

We find similar evidence when we measure *EDGAR* with $ESV\ Human_{i,t}$ (column 2) and $ESV\ Robot_{i,t}$ (column 3) and when we switch to $EC\ Size\ EAs_t$ (columns 3–6). Interestingly, estimated coefficients appear somewhat larger and more precisely estimated for nonrobot IP search intensity relative to robot IP search intensity, suggesting that investors demonstrate a greater susceptibility to the attention-constraining effects of earnings clusters and possibly use modern programmatic tools to alleviate these constraints.

We gauge the robustness of these findings in table 6, where we measure earnings clusters with $EC\ \#EAs_{i,t}$ in panel A and $EC\ Size\ EAs_{i,t}$ in panel B. Our results are broadly robust to (1) using $GFSVI_{i,t}$, the abnormal Google search intensity for firm i on its earnings announcement date t, as an alternative measure of investor information acquisition; (2) identifying bellwether firms with an alternative measure based on the correlation between detrended GDP and firms' sales; (3) including firm fixed effects to reduce the risk that our findings are driven by time-invariant firm-specific unobservable firm characteristics; and (4) estimating equation (3) using Poisson.[10] Our findings also hold if we use a measure of abnormal EDGAR searches, defined as firm i's *ESV* on its earnings announcement date minus the median *ESV* over the previous 10 weeks (Drake et al. 2020) (table 3 of the online appendix). In three of the eight columns, we observe not only that investors do not reduce their firm-specific information acquisition for bellwethers during earnings clusters, as in our main findings, but they even increase it, suggesting perhaps even stronger reallocation of attention to macroeconomic information. Note, however, that, in these analyses, we fail to find a reduction in information acquisition during earnings clusters, even for nonbellwether firms. Finally, in untabulated analyses, we find that our results are robust to using measures of earnings clusters constructed using either the unrestricted Compustat sample or taking a firm's announcement date as the earlier of Compustat or I/B/E/S.

[10] We estimate (pseudo-)Poisson regression models with multiple high-dimensional fixed effects using the Stata command *ppmlhdfe* from Correia, Guimarães, and Zylkin 2020.

TABLE 6

Investor Attention to Bellwethers and Earnings Clusters—Robustness

$$Y_{i,t} = \alpha + \beta_1 \text{ EC}_t \times \text{ Bell}_{i,t} + \beta_2 \text{ EC}_t + \beta_3 \text{ Bell}_{i,t} + \sum \gamma_j \text{ Controls}_{j,i,t} + \eta_w + \epsilon_t$$

Panel A: EC_t = *EC #EAs*$_t$

$Y_{i,t}$ =	$GSVI_{i,t}$	$ESV_{i,t}$		
	(1)	(2)	(3)	(4)
$EC_t \times Bell_{i,t}$	0.279**	0.147***	0.160***	0.057
	(2.08)	(3.92)	(3.58)	(0.81)
EC_t	−0.013	−0.227***	−0.250***	−0.327***
	(−0.17)	(−3.06)	(−3.50)	(−4.83)
$Bell_{i,t}$	−0.262**	−0.131***	−0.121***	−0.125**
	(−2.16)	(−3.86)	(−3.16)	(−2.20)
p-value of $\beta_1 + \beta_2 = 0$:	0.09	0.12	0.00	0.12
Controls	Yes	Yes	Yes	Yes
Controls × EC_t	Yes	Yes	Yes	Yes
Firm FEs	No	Yes	No	No
Year-Quarter FEs	Yes	Yes	Yes	Yes
Adj./Pseudo R^2	0.12	0.55	0.66	0.72
N Firm-Quarters	24,534	100,485	100,327	100,485

Panel B: EC_t = *EC Size EAs*$_t$

$Y_{i,t}$ =	$GSVI_{i,t}$	$ESV_{i,t}$		
	(1)	(2)	(3)	(4)
$EC_t \times Bell_{i,t}$	0.373***	0.097***	0.086**	0.130**
	(3.14)	(2.80)	(2.07)	(2.01)
EC_t	−0.044	−0.013	−0.047	−0.102
	(−0.54)	(−0.24)	(−0.92)	(−1.29)
$Bell_{i,t}$	−0.333***	−0.086***	−0.055	−0.182***
	(−3.06)	(−2.95)	(−1.57)	(−3.89)
p-value of $\beta_1 + \beta_2 = 0$:	0.01	0.39	0.74	0.05
Controls	Yes	Yes	Yes	Yes
Controls × EC_t	Yes	Yes	Yes	Yes
Firm FEs	No	Yes	No	No
Year-Quarter FEs	Yes	Yes	Yes	Yes
Adj./Pseudo R^2	0.12	0.55	0.66	0.72
N Firm-Quarters	24,534	100,485	100,327	100,485

Estimation: Fixed-effects (within) estimation in all columns excluding (Poisson). Year-week fixed effects and firm fixed effects are as indicated. Standard errors are clustered by year-quarter. All continuous variables are standardized. Statistical significance is denoted as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (two-tailed), t statistics are in parentheses. Intercept suppressed. **Dependent Variable**: $Y_{i,t}$ is either $GFSVI_{i,t}$, Google search intensity for firm i's ticker symbol; or $ESV_{i,t}$, the EDGAR search intensity for firm i on its earnings announcement day t. **Variable of Interest**: EC_t is either *EC #EAs*$_t$, the decile rank of the number of firms announcing earnings on day t, or *EC Size EAs*$_t$, the decile rank of the size-weighted number of firms announcing earnings on day t (both re-scaled to fall between (0,1)). $Bell_{i,t}$ is an indicator for firm i being designated a bellwether. **Control Variables**: Absolute returns ($|Ret|_{i,t}$), book-to-market ratio ($B/M_{i,t}$), financial leverage ($Leverage_{i,t}$), firm size ($Size_{i,t}$), analyst following ($\#Analysts_{i,t}$), bid-ask spread ($Spread_{i,t}$), trading volume ($Volume_{i,t}$), news coverage (*News Stories*$_{i,t}$), institutional ownership (*Inst Own*$_{i,t}$), an indicator for a firm's fourth-quarter EA (*Quarter 4*$_{i,t}$), and an indicator identifying days in which monetary policy, labor, inflation, or GDP information is released (*Macro Announce*$_t$). Each control is also interacted with EC_t.

4.2 MACRO CONTENT OF EARNINGS CONFERENCE CALLS DURING EARNINGS CLUSTERS

The findings in table 5 show that investors acquire less information about announcing firms during earnings clusters but that this reduction in information acquisition is muted for bellwethers. We interpret these findings as suggesting that investors' increased macro information acquisition during earnings clusters influences their acquisition and processing of firm-specific information on those days at the extensive margin. Namely, it changes which firms they pay attention to.

In the next analysis, we study whether the macro information acquisition patterns we document also change investors' acquisition of firm-specific information at the intensive margin. That is, conditional on paying attention to a firm, do investors change the information they collect about that firm's performance during earnings clusters? We argue that, with their commitment to gathering information on macroeconomic factors during earnings clusters, investors are more likely to ask macro-related questions in the conference calls hosted by bellwethers relative to nonbellwethers on these days.

To test this prediction, we re-estimate equation (3) with two modifications. First, we estimate the regression on the subsample of firm earnings announcements for which we have earnings conference call transcripts. Because by definition we are selecting on investors that participate in the earnings conference call, this sample holds the extensive margin investor choice constant. Second, we use percentage of macro-oriented words included in the transcript of firm i's earnings announcement conference call on day t ($CC\ \%_{i,t}$) as the outcome variable: If investors pay more attention to macro information during earnings clusters, they should induce relatively more macro discussions in earnings conference calls held by bellwethers on these days. We calculate the percentage of macro words in a transcript by counting the incidence of macro-oriented words and dividing by the total words.[11] The coefficient of interest is again β_1, which describes how the relation between intensity of macro words included in earnings conference calls and earnings clusters varies for bellwethers versus nonbellwethers. We expect the coefficient to be positive if our hypothesis holds in the data.

Table 7, which presents parameter estimates, shows that the percentage of macro words in earnings conference calls hosted by bellwethers during earnings clusters is relatively larger than the percentage in conference calls hosted by nonbellwethers on the same day ($\beta_1 > 0$ at the 10% significance level). The effects are economically meaningful. The difference in macro words in the conference calls of bellwethers versus nonbellwethers increases by 0.15 to 0.16 standard deviations as we move from the bottom to the top decile of EC_t (columns 1 and 4), amounting to 0.7 percentage

[11] Examples of macro words include "gross domestic product," "GDP," "exchange rate," "manufacturing activity," "monetary policy," "unemployment rate," "industrial production," and "inflation." Details on our macro dictionary are reported in appendix B.

TABLE 7
Investor Attention to Macro Information in Conference Calls and Earnings Clusters

$$\text{Macro Acq CC}_{i,t} = \alpha + \beta_1 \text{ EC}_t \times \text{Bell}_{i,t} + \beta_2 \text{ EC}_t + \beta_3 \text{ Bell}_{i,t} + \sum \gamma_j \text{ Controls}_{j,i,t} + \psi_w + \epsilon_{i,t}$$

$\text{EC}_t =$	*EC #EAs$_t$*			*EC Size EAs$_t$*		
	(*CC* %)	(*CC Present* %)	(*CC Q&A* %)	(*CC* %)	(*CC Present* %)	(*CC Q&A* %)
$EC_t \times Bell_{i,t}$	0.158*	0.039	0.228***	0.147*	0.075	0.159**
	(1.87)	(0.40)	(3.55)	(1.79)	(0.76)	(2.29)
EC_t	−0.218***	−0.224***	−0.179***	−0.185***	−0.198***	−0.138***
	(−4.15)	(−4.02)	(−3.34)	(−4.45)	(−4.43)	(−3.38)
$Bell_{i,t}$	−0.072	0.040	−0.159***	−0.057	0.012	−0.094*
	(−0.97)	(0.46)	(−3.01)	(−0.80)	(0.14)	(−1.74)
$\lvert Ret\rvert_{i,t}$	0.032*	0.048**	0.023*	0.019	0.021	0.027**
	(1.99)	(2.58)	(1.86)	(1.45)	(1.33)	(2.55)
$B/M_{i,t}$	−0.098***	−0.071***	−0.079***	−0.047*	−0.029	−0.047*
	(−4.77)	(−3.55)	(−3.70)	(−2.02)	(−1.38)	(−1.87)
$Leverage_{i,t}$	0.101***	0.106***	0.079***	0.083***	0.075***	0.067***
	(3.34)	(3.57)	(2.82)	(4.49)	(4.26)	(3.29)
$Size_{i,t}$	0.213***	0.182***	0.200***	0.188***	0.144***	0.181***
	(6.74)	(6.02)	(5.94)	(5.92)	(5.63)	(4.97)
$\#Analysts_{i,t}$	−0.093***	−0.041*	−0.083***	−0.089***	−0.051**	−0.072***
	(−4.53)	(−1.83)	(−4.19)	(−5.24)	(−2.39)	(−4.07)
$Spread_{i,t}$	−0.003	−0.014	0.003	0.004	−0.004	−0.001
	(−0.08)	(−0.29)	(0.06)	(0.13)	(−0.11)	(−0.03)
$Volume_{i,t}$	−0.076***	−0.071***	−0.058***	−0.080***	−0.069***	−0.070***
	(−6.11)	(−5.62)	(−4.96)	(−5.59)	(−5.34)	(−5.34)
$News\ Stories_{i,t}$	0.050**	0.032	0.041*	0.090***	0.098***	0.053**
	(2.21)	(1.33)	(1.99)	(4.48)	(4.61)	(2.36)
$Inst\ Own_{i,t}$	0.097***	0.070***	0.095***	0.086***	0.063***	0.088***
	(5.91)	(4.40)	(4.93)	(5.79)	(5.90)	(4.70)
$Quarter\ 4_{i,t}$	0.004	−0.050	0.040	−0.046	−0.104**	0.021
	(0.06)	(−0.89)	(0.68)	(−0.92)	(−2.24)	(0.44)
$Macro\ Announce_t$	0.027	0.076*	−0.031	0.025	0.049	−0.024
	(0.74)	(1.76)	(−0.72)	(0.63)	(1.16)	(−0.49)
p-value of $\beta_1 + \beta_2 = 0$:	0.51	0.11	0.53	0.66	0.25	0.79
p-value of $(\beta_1^{\text{Present}} - \beta_1^{\text{Q\&A}})$:		0.05			0.16	
Controls × EC_t	Yes	Yes	Yes	Yes	Yes	Yes
Year-Week FEs	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.07	0.06	0.04	0.07	0.06	0.04
N Firm-Quarters	53,247	53,247	53,247	53,247	53,247	53,247

Estimation: Fixed-effects (within) estimation with year-week fixed effects. Standard errors are clustered by year-quarter. All continuous variables are standardized. Statistical significance is denoted as follows: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (two-tailed), t statistics are in parentheses. Intercept suppressed. The tests for equality of coefficients ($\beta_1^{\text{Present}} - \beta_1^{\text{Q\&A}}$) is one-tailed. **Dependent Variable**: *Macro Acq* $CC_{i,t}$ is the intensity of macro words in firm i's entire conference call, presentation segment, and question and answer segment *CC* $\%_t$, *CC Present* $\%_t$, or *CC Q&A* $\%_t$, respectively. **Variable of Interest**: EC_t is either *EC #EAs$_t$*, the decile rank of the number of firms announcing earnings on day t, or *EC Size EAs$_t$*, the decile rank of the size-weighted number of firms announcing earnings on day t (both re-scaled to fall between (0,1)). $Bell_{i,t}$ is an indicator for firm i being designated a bellwether. **Control Variables**: Absolute returns ($\lvert Ret\rvert_{i,t}$), book-to-market ratio ($B/M_{i,t}$), financial leverage ($Leverage_{i,t}$), firm size ($Size_{i,t}$), analyst following ($\#Analysts_{i,t}$), bid-ask spread ($Spread_{i,t}$), trading volume ($Volume_{i,t}$), news coverage ($News\ Stories_{i,t}$), and institutional ownership

(*Continued*)

TABLE 7—*(Continued)*

($Inst\ Own_{i,t}$), an indicator for a firm's fourth-quarter EA ($Quarter\ 4_{i,t}$), and an indicator identifying days in which monetary policy, labor, inflation, or GDP information is released ($Macro\ Announce_t$). Each control is also interacted with EC_t.

points (or a 29% increase relative to the unconditional sample mean).[12] Furthermore, while the estimated β_2 coefficient is negative and significant, indicating that the macro content of earnings conference calls hosted by nonbellwether firms declines during earnings clusters, the sum of β_1 and β_2 is statistically insignificant from zero. Thus, bellwethers do not experience a decline in the macro content of earnings conference calls, even when investors' attention capacity is strained.

Our findings can manifest in at least two ways. First and foremost, investors and analysts could express a revealed preference for macro information by asking more macro-oriented questions during the Q&A portion of the earnings call. Second, managers may anticipate this increase in demand for macro information during earnings clusters and preemptively attempt to provide more macro information during their presentation. We remain agnostic as to which of these, if either, predominate and assess empirically whether the relative increase in the intensity of macro information we document is driven by the presentation or Q&A portion of the call. We decompose $CC\ \%_{i,t}$ into the percentage of macro words in the presentation portion ($CC\ Present\ \%_{i,t}$) and the percentage of macro words in the Q&A portion ($CC\ Q\&A\ \%_{i,t}$). We find that the results are driven entirely by the Q&A. The estimated β_1 coefficient is positive but not statistically significant for $CC\ Present\ \%_{i,t}$ and positive and statistically different from zero at the 1% level for $CC\ Q\&A\ \%_{i,t}$. The difference in macro words included in the Q&As of bellwethers versus nonbellwethers increases between 0.23 and 0.16 standard deviations, as we move from the bottom to the top decile of *EC* (columns 3 and 6), which, in economic terms, amounts to 1.03 percentage points (or a 55% increase relative to the unconditional sample mean). The difference between the two coefficient estimates for $CC\ Q\&A\ \%_{i,t}$ and $CC\ Present\ \%_{i,t}$ is positive, large, and statistically significant for $EC\ \#EAs_t$ (p-value = 0.05) but not for $EC\ Size\ EAs_t$ (p-value = 0.16). Overall, this is consistent with investors demanding more macro information from bellwethers during earnings clusters and not with managers preemptively disclosing this information.

Jointly considered, the evidence in tables 5 and 7 strongly supports the notion that investors' increased acquisition of macro information during earnings clusters affects their acquisition of firm-specific information on those same days at both the extensive and intensive margins.

[12] The raw and within-fixed-effect standard deviation is almost identical.

FIG. 1.—Information acquisition, earnings clusters, and capital market outcomes.

4.3 EQUITY VALUATIONS AND MACRO AND FIRM INFORMATION ACQUISITION DURING EARNINGS CLUSTERS

Our findings that (1) investors acquire more macro information during earnings clusters and that (2) this pattern influences the processing of firm-specific information has implications for the incorporation of information into stock prices, as illustrated in figure 1.

To study these implications, we start with an analysis of whether the Hirshleifer, Lim, and Teoh 2009 finding that earnings news is priced less efficiently during earnings clusters is muted for bellwethers because they do not experience as much of a reduction in attention during earnings clusters, as we documented in the previous two subsections. Specifically, we estimate the following regression on a sample constructed at the firm-earnings announcement level:

$$\begin{aligned} \mathrm{Y}_{i,t} &= \alpha + \beta_1\,\mathrm{EC}_t \times \mathrm{Bell}_{i,t} \times \mathit{SUE}_{i,t} + \beta_2\,\mathrm{EC}_t \times \mathrm{Bell}_{i,t} + \beta_3\,\mathrm{EC}_t \times \mathit{SUE}_{i,t} \\ &+ \beta_4\,\mathrm{Bell}_{i,t} \times \mathit{SUE}_{i,t} + \beta_5\,\mathrm{EC}_t + \beta_6\,\mathrm{Bell}_{i,t} + \beta_7 \mathit{SUE}_{i,t} + \sum \gamma_j\,\mathrm{Controls}_{j,i,t} + \psi_w + \epsilon_{i,t}. \quad (4) \end{aligned}$$

The dependent variable is either the cumulative abnormal return over the two-trading-day window [0,1] around the earnings announcement ($CAR_{i,[01]}$) or the cumulative raw return over the two-trading-day window [0,1] around the earnings announcement ($Ret_{i,[01]}$). EC_t and $Bell_{i,t}$ are defined as before, while $SUE_{i,t}$ is a variable that assigns firms' earnings surprises to 11 quantiles, where firms with negative surprises are equally assigned to quantiles one to five, firms with zero surprises are assigned to quantile six, while firms with positive surprises are equally assigned to quantiles seven to eleven (Hirshleifer and Sheng 2022-09).[13] We include the same set of controls, control interactions, and fixed effects (year-week) discussed for equation (3) and cluster standard errors in the same manner (at the year-quarter level). We further expand the set of controls to include $Beta_{i,t}$, calculated using daily return during the previous three months, and the interaction between $Beta_{i,t}$, $B/M_{i,t}$, $Leverage_{i,t}$ and $Size_{i,t}$ with $SUE_{i,t}$ and $EC_t \times SUE_{i,t}$ because prior work shows that ERCs are a function of the growth, riskiness, and size of the firm (Collins and Kothari 1989, Easton and Zmijewski 1989, Gipper, Leuz, and Maffett 2020).

[13] In this analysis, we measure EC_t with *EC #EAs*$_t$. Our inferences are similar with *EC Size EAs*$_t$ (table 4 of the online appendix).

There are three coefficients of interest to us. The first two are β_7 and β_3. We expect investors to respond more strongly to firms announcing more positive earnings surprises, also known as the earnings response coefficient ($\beta_7 > 0$), but we expect this effect to be muted on earnings cluster days ($\beta_3 < 0$), as shown by Hirshleifer, Lim, and Teoh 2009. The third coefficient of interest is β_1: We expect the relation between the earnings response coefficient and earnings clusters to weaken (i.e., be less negative) for bellwethers, because they do not experience as much of a reduction in attention during earnings clusters ($\beta_1 > 0$). Estimated coefficients in table 8, column 1 (for $CAR_{i,[01]}$) and column 4 (for $Ret_{i,[01]}$) comport with our expectations. We confirm a positive stock market response to earnings surprises that is muted during earnings clusters: Moving from the bottom to the top decile of the earnings cluster distribution reduces the earnings response coefficient by approximately 42%. Importantly, we also find that the muting effect of earnings clusters is attenuated for bellwethers, consistent with investors reducing their attention to these firms to a lesser extent during earnings clusters. Moreover, the effect is economically meaningful, as it indicates that bellwethers' earnings response coefficients do not decline during earnings clusters. ($\beta_1 + \beta_3$ is statistically indistinguishable from zero, with a p-value of 0.80.)

We next explore the mediation role of investors' macro information acquisition for these findings, as illustrated in figure 1. If investors can more easily interpret bellwether earnings news, due to their increased collection of macroeconomic information during earnings clusters, as we argue, then the market response to earnings surprises for bellwethers during earnings clusters should strengthen on days with extensive collection of macroeconomic information. This intuition resembles that of Hirshleifer and Sheng 2022-09, who emphasize that macro announcements and corporate earnings can complement each other, and investors may rely more on corporate earnings to gain macro insights on macro announcement days.

To test this prediction, we repeat our equity valuation analysis separately on days with high (above-median *MGSVI*) and low (below-median *MGSVI*) macro information acquisition (*High MGSVI* and *Low MGSVI*, respectively). In columns 2 and 3, where the outcome variable is $CAR_{i,[01]}$, we find that the estimated β_1 coefficient is positive, large, and statistically different from zero (at the 1% level) on days with high macro information acquisition, while it is positive but small and statistically indistinguishable from zero on days with low macro information acquisition. The difference between the two coefficient estimates is statistically significant (p-value = 0.02), consistent with our findings in column 1 arising from days with high macro information acquisition. We document similar evidence in columns 5 and 6, where the outcome variable is $Ret_{i,[01]}$: The estimated β_1 coefficient is positive, large, and statistically different from zero (at the 1% level) on days with high macro information acquisition, while it is positive but smaller and statistically indistinguishable from zero on days with low macro information acquisition, and the difference between the two is statistically significant

TABLE 8
The Mediating Effect of Information Acquisition on Firm ERCs

$$Y_{i,t} = \alpha + \beta_1\,\mathrm{EC}_t \times \mathrm{Bell}_{i,t} \times SUE_{i,t} + \beta_2\,\mathrm{EC}_t \times \mathrm{Bell}_{i,t} + \beta_3\,\mathrm{EC}_t \times SUE_{i,t} + \beta_4\,\mathrm{Bell}_{i,t} \times SUE_{i,t} + \beta_5\,\mathrm{EC}_t + \beta_6\,\mathrm{Bell}_{i,t} + \beta_7 SUE_{i,t} + \sum \gamma_j\,\mathrm{Controls}_{j,i,t} + \psi_w + \epsilon_{i,t}$$

$Y_{i,t} =$	$CAR_{i,[01]}$			$Ret_{i,[01]}$		
	(All)	(High MGSVI)	(Low MGSVI)	(All)	(High MGSVI)	(Low MGSVI)
$EC_t \times Bell_{i,t} \times SUE_{i,t}$	0.718***	1.730***	0.254	0.958***	2.065***	0.591
	(2.70)	(2.80)	(0.82)	(3.56)	(2.95)	(1.33)
$EC_t \times Bell_{i,t}$	−0.206	−0.567	−0.090	−0.262	−0.592	−0.239
	(−0.77)	(−1.45)	(−0.23)	(−0.98)	(−1.35)	(−0.58)
$EC_t \times SUE_{i,t}$	−0.777***	−0.935***	−0.614***	−0.974***	−1.108***	−0.835***
	(−6.74)	(−7.37)	(−4.05)	(−7.53)	(−7.22)	(−4.56)
$Bell_{i,t} \times SUE_{i,t}$	−0.532**	−1.553***	−0.042	−0.726***	−1.836***	−0.319
	(−2.09)	(−2.72)	(−0.16)	(−2.84)	(−2.84)	(−0.88)
EC_t	−0.330	−0.423	−0.311	−0.794*	−0.860	−1.074
	(−1.27)	(−1.19)	(−0.65)	(−1.92)	(−1.41)	(−1.62)
$Bell_{i,t}$	0.269	0.532	0.197	0.309	0.572	0.311
	(1.21)	(1.52)	(0.61)	(1.36)	(1.47)	(0.87)
$SUE_{i,t}$	1.837***	1.897***	1.802***	2.084***	2.128***	2.066***
	(17.63)	(15.22)	(13.84)	(17.42)	(14.14)	(12.77)
p-value of $\beta_1 = \beta_3$:	0.80			0.95		
p-value of $\beta_1^{\mathrm{High}} - \beta_1^{\mathrm{Low}}$:		0.02			0.04	
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Controls × EC_t	Yes	Yes	Yes	Yes	Yes	Yes
Year-Week FEs:	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.07	0.07	0.08	0.11	0.12	0.14
N Firm-Quarters	138,224	78,268	53,235	138,224	78,268	53,235

Estimation: Fixed-effects (within) estimation with year-week fixed effects. Standard errors are clustered by year-quarter. All continuous variables are standardized. Statistical significance is denoted as follows: $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (two-tailed), t statistics are in parentheses. Intercept suppressed. *High (Low) MGSVI* are for days having an above (below)-median *MGSVI* search intensity. **Dependent Variable**: $Y_{i,t}$ is either $CAR_{i,[01]}$, firm i's abnormal returns in the day of and following the earnings announcement; or $Ret_{i,[01]}$, firm i's raw returns in the day of and following the earnings announcement. **Variable of Interest**: EC_t is *EC #EAs*$_t$. $Bell_{i,t}$ is an indicator for firm i being designated a bellwether. $SUE_{i,t}$ is a variable that assigns firms' earnings surprises to 11 quantiles. **Control Variables**: Absolute returns ($|Ret|_{i,t}$), stock beta ($Beta_{i,t}$), book-to-market ratio ($B/M_{i,t}$), financial leverage ($Leverage_{i,t}$), firm size ($Size_{i,t}$), analyst following ($\#Analysts_{i,t}$), bid-ask spread ($Spread_{i,t}$), trading volume ($Volume_{i,t}$), news coverage ($News\ Stories_{i,t}$), and institutional ownership ($Inst\ Own_{i,t}$), an indicator for a firm's fourth-quarter EA ($Quarter\ 4_{i,t}$), and an indicator identifying days in which monetary policy, labor, inflation, or GDP information is released ($Macro\ Announce_t$). Each control is also interacted with EC_t. We further interact important ERC determinants (*ERC Det.*), namely $Beta_{i,t}$, $B/M_{i,t}$, $Leverage_{i,t}$ and $Size_{i,t}$, with $SUE_{i,t}$ and $EC_t \times SUE_{i,t}$.

(p-value = 0.04). Overall these results are consistent with the estimated effect of bellwethers operating through investor macro information acquisition and therefore with our argument that investors achieve economies of scale in their information acquisition during earnings clusters.

These analyses suggest that investor macro information acquisition contributes to the underreaction to earnings information documented in the literature (e.g., Hirshleifer, Lim, and Teoh 2009) and that investors do not lose their ability to pay attention during earnings clusters; rather, they shift their attention to interpreting macro implications and adjust their attention to firm information accordingly to achieve economies of scale.

5. *Conclusion*

We study investors' acquisition of macroeconomic information during earnings clusters. We predict that investors shift their attention to macroeconomic factors and therefore process more macro information during earnings clusters. This, in turn, can explain why we observe less processing of firm-specific information by investors during earnings clusters. We offer evidence consistent with this prediction. Specifically, we document that (1) investors acquire more macro information during earnings clusters, (2) this behavior is amplified during negative economic shocks and concurrent macro announcements, and (3) this information acquisition has important implications for firm-specific information acquisition and equity valuations. We also show that these findings are likely driven by the fact that earnings announced during clusters are more informative about the macroeconomy and that investors try to economize their information acquisition due to attention constraints.

Our results provide an additional explanation for prior findings that investors gather and process less firm-specific information on earnings cluster days. In contrast to the literature, which has interpreted this empirical regularity as evidence that investors become distracted, our paper suggests that investors efficiently reallocate their attention to the most beneficial information. Our findings indicate that the contemporaneous release of microeconomic information by multiple firms stimulates the acquisition of macro information. This insight deepens the understanding of the link between the macroeconomy and asset prices and, in particular, of the drivers of investors' macro information acquisition. Moreover, we show that investors use earnings news to update their priors not only about the future performance of the firm but also about the economy. This is important because our evidence lends empirical support to a mechanism that has been largely assumed by influential recent research in accounting and finance.

Our paper has limitations. First, investors' attention allocation and information acquisition are unobservable. While we rely on multiple approaches and triangulate our findings using predictions from theory, we can only infer investors' behavior using imperfect observable measures. Second, we do not have exogenous variation in earnings clusters. Thus, although most changes to firms' announcement dates are benign (DeHaan, Shevlin, and Thornock 2015 p. 39), our evidence could reflect firms announcing earnings strategically when investors have more appetite for more macro information or to hide negative firm-specific news. Readers should keep in mind these caveats when interpreting our findings.

Appendices

APPENDIX A: VARIABLE DEFINITIONS AND DATA SOURCES

Variable	Definition	Source
Agg ROA (ROE)$_t$	Aggregate ROA or ROE for firms announcing earnings during earnings clusters on day t. ROA (ROE) is calculated as the sum of earnings (*ibq*) deflated by the sum of lagged assets or equity (*atq* or *ceqq*).	Compustat
Agg ROA (ROE) Cluster (Non-Cluster)$_t$	Aggregate ROA or ROE for firms announcing earnings during earnings clusters in quarter t. ROA (ROE) is calculated as the sum of earnings (*ibq*) deflated by the sum of lagged assets or equity (*atq* or *ceqq*). Clusters (Non-Clusters) are determined as being above (below) the median number of daily earnings announcements in quarter t.	Compustat
#Analysts$_{i,t}$	The number of analysts following firm i in the quarter for which earnings are announced on day t.	I/B/E/S
B/M$_{i,t}$	The book value of common equity (Compustat variable *ceq*) scaled by the market value of common equity (Compustat variables *prccq*× *cshoq*), at the beginning of quarter t.	Compustat
Bad$_t$	Indicator for bad economic states in day t. Bad state is a larger than two standard deviation negative revision in economic conditions on day t using the Aruoba-Diebold-Scotti Business Condition Index (Aruoba, Diebold, and Scotti 2009).	Federal Reserve Bank of Philadelphia
Bell$_{i,t}$	An indicator for firm i being a bellwether in quarter t. Bellwether firms are determined by being above the 90th percentile of the absolute correlation of firm i's previous five quarters' earnings stream with GDP detrended using the Baxter and King 1999 bandpass filter.	Compustat/St. Louis Fed
Beta	The stock beta for firm i at the end of quarter q, calculated using daily stock returns over the previous 90 days.	Compustat/CRSP
CC %	The percent of macro words in firm i's conference call on day t. Our macro dictionary is described in appendix B	Capital IQ
CC Present %	The percent of macro words in the presentation portion of firm i's conference call on day t.	Capital IQ
CC Q&A %	The percent of macro words in the Q&A portion of firm i's on day t.	Capital IQ

Variable	Definition	Source
#EAs$_t$	The number of earnings announcements released during day t. Dates in the Compustat Quarterly database must agree with the date from IBES.	Compustat/ IBES

Variable	Definition	Source
EC #EAs$_t$	The year-quarter decile distribution of the number of earnings announcements released during day t, rescaled to fall between (0,1). Dates in the Compustat Quarterly database must agree with the date from IBES.	Compustat/ IBES
EC Size EAs$_t$	The year-quarter decile distribution of the sum of total assets (Compustat variable *atq*) for all firms that announce earnings during time t, divided by the sum of total assets for all firms in the sample during the previous quarter.	Compustat
ESV$_{i,t}$	The total EDGAR downloads for firm i's 10-K and 10-Q filings on day t.	SEC EDGAR
ESV Human$_{i,t}$	*ESV*$_{i,t}$ for IP addresses not labeled 'Robot'.	SEC EDGAR
ESV Robot$_{i,t}$	*ESV*$_{i,t}$ for IP addresses labeled 'Robot' (more than 50 downloads per minute).	SEC EDGAR
GFSVI$_{i,t}$	Firm i's Google search intensity for day t, minus its average search intensity over the prior eight weeks.	Google Trends
Inst Own$_{i,t}$	Firm i's percent of institutional ownership on day t.	Thompson Reuters
Leverage$_{i,t}$	Firm i's leverage for the quarter for which earnings are announced on day t, calculated as the ratio of long-term debt to total assets (Compustat variables *dlttq* / *atq*), prior to forecast issuance.	Compustat
Macro Announce$_t$	An indicator identifying days in which monetary policy, labor (unemployment and jobs updates), inflation, or GDP information is released: Monetary policy announcement dates are taken from the Federal Reserve's calendar. Labor announcement dates and inflation announcement dates are taken from the BLS release calendar. GDP (the "advance" estimate, which is the first official estimate of GDP made by the BEA) announcement dates are taken from the BEA's release calendar.	Federal Reserve , BLS , BEA
Market Ret$_t$	The value-weighted market return for day t.	CRSP
MGSVI$_t$	Google search intensity for all macro terms on day t. Calculated as the sum of Google SVIs for all macro search terms (see appendix B for details on macro search terms) on day t.	Google Trends

Variable	Definition	Source
News Stories$_{i,t}$	The number of news stories about firm i published on day t.	Ravenpack
NGDP (CPI) Growth$_t$	is the growth in nominal GDP (CPI) during quarter t. We use the "Final" vintage for GDP.	FRED Database
Quarter 4$_{i,t}$	An indicator for firm i announcing its 4th quarter earnings on day t.	Compustat
$\lvert Ret_{i,t} \rvert$	Firm i's absolute return on day t.	CRSP

Variable	Definition	Source
$Size_{i,t}$	Firm i's size at the beginning of the quarter for which earnings are announced on day t, calculated as the natural logarithm of firm i's assets at the beginning of the quarter (Compustat variable *atq*).	Compustat
$Spread_{i,t}$	Firm i's bid-ask spread calculated with the CRSP variables *bid* - *ask* / *prc*.	CRSP
$SUE_{i,t}$	The rank of firms' earnings surprises. We assign earnings surprises in 11 quantiles for each year. Firms with negative surprises are equally assigned to quantiles 1 to 5, and firms with positive surprises are equally assigned to quantiles 7 to 11. Firms with zero surprises are labeled as quantile 6. Earnings surprises are calculated as firm i's actual EPS minus its mean consensus EPS forecast prior to the earnings announcement date, scaled by its stock price at the end of the quarter.	IBES
$Volume_{i,t}$	The volume of trades for firm i on day t, divided by firm i's total shares outstanding on day t.	CRSP

APPENDIX B: MACRO DICTIONARY

Our conference call and Google macro search intensity tests are based on the following search terms.

B.1 DICTIONARY TERMS

Aggregate consumption, aggregate demand, aggregate investment, aggregate production, economic activity, economic forecast, economic outlook, gross domestic product, gdp, exchange rate, deficit, interest rate, manufacturing activity, monetary policy, unemployment rate, industrial production, inflation, production capacity, unemployment, jobs numbers, recession, bear market, stock crash, financial crisis, yield curve, inverted yield, consumer spending, fed announcement, fomc, macroeconomy, factors of production, bull market, pet food, oil change, nfl, nba, and mlb.[14]

B.2 GOOGLE TRENDS EXTRACTION

Because Google deflates search volume by the maximum search in a given search time frame and does not provide daily time series longer than nine months at a time, we extract the daily data for each search term in one-month increments, then adjust each daily series by its monthly SVI value, yielding time series that are comparable from start to finish. We then form category indices of search by taking averages of the SVIs for each variable in the category, as follows:

[14] Note: Google search terms are not case sensitive. See https://support.google.com/websearch/answer/134479?hl=en&rd=1.

- ***MGSVI Demand***: *aggregate consumption, aggregate demand, and consumer spending.*
- ***MGSVI Supply***: *aggregate investment, aggregate production, manufacturing activity, industrial production, and production capacity.*
- ***MGSVI Outcome***: *economic activity, economic forecast, economic outlook, gdp, gross domestic product, jobs numbers, recession, and unemployment rate.*
- ***MGSVI Policy***: *interest rate, monetary policy, yield curve, inverted yield, fed announcement, and fomc.*
- ***MGSVI Equity***: *bear market, bull market, stock crash, and financial crisis.*
- ***Non-Macro GSVI***: *pet food, oil change, nfl, nba, and mlb.*

C Examples of Macro Topics in Earnings Conference Call Q&As

C.1 POTBELLY Q&A, MAY 3, 2016

C.1.1 Question.

C.1.1.1 Gregory Ryan Francfort, BofA Merrill Lynch, Research Division. On the **commodity inflation, what was commodity inflation in the quarter?** *And then, in the 28% to 29%, or I guess the bottom end of the 28% to 29% range.* **Does that suggest potentially deflation for the year?**

C.1.2 Answer.

C.1.2.1 Michael W. Coyne, Former Senior VP & CFO. We haven't said that. We said on the last call, we expected modest inflation. And we are still saying there's been – **there's a little bit of an inflation**, *a point or less in the first quarter. There are certain categories for sure where we're getting the benefit of some deflation.* **But overall, still what I call very modest inflation going forward**.

C.2 EATON CORPORATION Q&A, OCTOBER 20, 2008

C.2.1 Question.

C.2.1.1 Andrew Obin, Merrill Lynch. What are your **assumptions for GDP growth** *for the US and the world in 4Q and '09, just because your forecast has to be reliant on the macro work that you guys are so good at?*

C.2.2 Answer.

C.2.2.1 Richard H. Fearon, Executive Vice President and CFO. Our current view is that the **US GDP growth in 4Q we think will be -2.0% and with an industrial production number that's more negative than that**.

C.2.3 Answer.

C.2.3.1 Alexander M. Cutler, Chairman and CEO. I think that's the real significance because I think as you'll all recall, it's generally when the US goes through a period of economic recession, **manufacturing tends to cycle more or has a larger beta if you will so a -2% in the overall GDP means that you get a fairly severe drop in terms of manufacturing industrial production**.

C.2.4 Follow-Up Question.

C.2.4.1 Andrew Obin, Merrill Lynch. What about ′ 09?

C.2.5 Answer.

C.2.5.1 Richard H. Fearon, Executive Vice President and CFO. Our overall view is that **GDP growth in ′09 for the US will be just about flat** *to perhaps slightly negative.*

C.2.6 Answer.

C.2.6.1 Alexander M. Cutler, Chairman and CEO. You start off with probably a negative in the first quarter and then you start to fight your way to being more flat as you get out of the second quarter.

C.2.7 Follow-Up Question.

C.2.7.1 Andrew Obin, Merrill Lynch. What assumptions are you making about **global growth?** *In particularly I'm very interested in your views on China?*

C.2.8 Answer.

C.2.8.1 Alexander M. Cutler, Chairman and CEO. As I mentioned earlier, our general view is that this economic period of weakness really triggered or magnified by the global credit crunch takes 2% to 2.5% off of basically every region of the world. We think that still means that your kind of **brick nations continue to be growing but they're growing at lower numbers, so China comes down from a 10% to 10.5% to probably something closer to 8.5%. Brazil comes down from kind of a 5% to 5.5% to something like 2.5% to 3%**. *We don't think that there's any area of the world that's going to escape this. Some will continue to grow; some will go to a flat or slightly negative status.*

C.3 KROGER Q&A, MARCH 3, 2011

C.3.1 Question.

C.3.1.1 Deborah L. Weinswig, Citigroup Inc, Research Division. **What do you think will be the impact of inflation on spending?** *And I'm sure you've done a lot of work on this with Rodney.*

C.3.2 Answer.

C.3.2.1 David B. Dillon, Former Chairman of the Board. It's going to be a little hard to read in advance, of course, because on one hand, you have some products that we know for sure, things like milk, when the retail price goes up, tonnage on that product has historically always gone down. But some other products don't seem to vary too much particularly if the price changes aren't significant. What we were particularly pleased with as we've begun to see some return of grocery inflation as opposed to the deflation we've been experiencing, we still saw in the fourth quarter solid tonnage growth, and of course, the identical sales were strong too. So I'd say I'm pretty bullish and optimistic about where this can lead. However, we're cautious because we do think we have a fragile economy, and we've seen many of our customers who are greatly affected by that. Rodney, you want to add anything?

C.3.2.2 Rodney McMullen. The only thing I would add is obviously that **the amount of inflation would affect that so if it's 3% to 4% overall it's one thing. If it happens to be 10% or above I think it's something different**. *The other thing that we believe and we see customers continuing to do this is as prices increase, our own corporate brand program offers an alternative. And at many times, that item will even be cheaper than where the customer was spending before when they were buying the national brand item. So we believe both of those things will help offset or minimize the impact. But obviously,* **the amount of inflation will drive that**, *in answer to your question, by a lot…* **And the only other thing I would add is if you look at everything out there, it looks like the inflation is going to be there for a while**. *So if you go back and look at a year or two ago, an awful lot of the things where the inflation was happening look like something that was really only maybe a few months in duration and it wasn't permanent. For now, it just looks much more permanent in terms of the inflation as you look out six to 12 months, you just see actually more of it coming rather than less of it coming. So I believe that's probably driving part of the behavior.*

C.3.3 Question.

C.3.3.1 John Edward Heinbockel, Guggenheim Securities, LLC, Research Division. **And how do you gauge or get comfortable with your customers' ability to absorb food inflation, gas?** *I mean it sounds funny, you walk through any retailer and so towels going up and appliances are going up because there's input costs there too. How do you kind of gauge that and remain sensitive to their ability to absorb it now. And then going forward, is it possible to get to a point where just less of it can be passed through later this year, next year just because that, particularly the low-income customer's under some stress?*

C.3.4 Answer.

C.3.4.1 David B. Dillon, Former Chairman of the Board. I think the answer really is a much like **what we found as we went into the recession**. *At first, customers make choices. They go through a rational way of thinking, how do I get through this. And sometimes they trade to different products, sometimes they trade to different retailers, sometimes they completely revise what they decide they're going to consume in their family. But the real advantage for us is the way we're able to get insight into segmented customer groups so that we can see that change of behavior. So for instance, the best illustration of this is that* **a number of our customers, mainstream and upscale kind of customers, are feeling a little better about the economy**. *We described that last quarter and it's still true today, things like Starbucks have improved within our stores. More said [ph] has improved, natural food is up strongly and is growing households as an example. So many of those areas continue just like we described before. But those are typically among customers who are feeling a little less affected by the economy. But there are many customers who are more price sensitive, who are greatly affected even still* **today, high unemployment and food stamps**, *illustrate that best. And the way we segment customers through our insight data allows us to see more clearly how that behavior changes. So we try to* *address that both with what we offer on pricing. We try to address that with what we do with our ads. We try to address it in what we offer in the stores.*

C.3.4.2 Rodney McMullen. The only other thing, Dave, I'd add to that, if you look there's also categories that we can take business away from, if you look at like restaurant. Now you can come to one of our stores and get a high-quality good-for-you meal at a third to half the price a restaurant would be, that really doesn't take any longer to prepare than what it takes for a customer to drive to a restaurant and get to their home, back to their home to eat it. So there's a lot of business out there to be moved from completely outside of our channel back into our channel too.

C.3.4.3 David B. Dillon, Former Chairman of the Board. And I think you also commented, with your comment about what happens with fuel prices and how does that affect behavior? That's important to note. That's one of the factors we have identified for this next year that could cause depending on, which way it goes, could cause a change in how we see the results either at the high end or low end of our results. And it illustrates why we think **there's some uncertainty out there that we've got to be a little cautious about**. *But we intend to be flexible, we intend to adjust as we go and we intend to read the market as we go through it in order to make the year come out strong.*

C.4 TEXAS ROADHOUSE Q&A, MAY 4, 2015

C.4.1 Question.

C.4.1.1 Joseph Terrence Buckley, BofA Merrill Lynch, Research Division. Two questions. Obviously, very, very strong same-store sales and most of your traffic. Are you seeing anything noteworthy in the way consumers are spending within the restaurants in terms of mix or alcohol or side items or anything of that sort that would suggest maybe **a little better consumer spending environment?**

C.4.2 Answer.

C.4.2.1 Wayne Kent Taylor, Founder, Chairman & CEO. Yes, this is Kent. No really – no real change in our sales mix… People are ordering the same items as they were before.

C.4.3 Follow-Up Question.

C.4.3.1 Joseph Terrence Buckley, BofA Merrill Lynch, Research Division. And then just one more question, just on labor. **Could you just talk wage rate inflation, sort of what you're seeing, or is there any signs of that begin to pick up as well?**

C.4.4 Answer.

C.4.4.1 Tonya R. Robinson, Chief Financial Officer. Right now, we continue to see about 1.5% to 2%, kind of in the range of what we were expecting. So we haven't seen anything more than we've expect so far this year.

C.4.4.2 Scott M. Colosi, President. There's no – Joe, this is Scott. I'd add one more thing to that. I mean, no doubt, we're seeing inflation pick up a little bit. We've seeing our turn over pick up a little bit as well, things that we would expect to see *with* **unemployment continuing to tick down. And I think unemployment, by itself, is probably one of the biggest drivers of sales in our industry**. *Just by having that benefit, it's very highly correlated to our sales momentum over the years, to the good or to the bad. And right now, of course, it's to the good, as* **we see unemployment continue to drop**.

C.4.5 Question.

C.4.5.1 Jeffrey Andrew Bernstein, Barclays Bank PLC, Research Division. Two questions. One is just a follow-up on that. When you think about beef, I think you guys are now refraining from sharing maybe what percentage is locked versus floating. But if not the case, by all means, love to know what percentage is locked versus floating. But otherwise, **what are you hearing from your suppliers in terms of the outlook going into next year?** *It seems like we're hearing from some that confidence is building.* **You could actually see whether it's inflation neutralized or actually turn favorable year-on-year when you're look into '16? Or what are you hearing thus far?**

C.4.6 Answer.

C.4.6.1 Scott M. Colosi, President. Jeff, this is Scott. I would tell you that at the end of the day, we really don't know. So there's so many variables that go into the price for beef well beyond just sort of corn and supply of cattle that we don't know – we don't think it'll be substantially – or prices will be substantially higher than this year. But of course, we've all been wanting to be able to know we're going to get relief in pricing. And we're still hopeful that eventually, and the key word is eventually, there will be relief in cattle prices. And certainly some of that will be supported by continued lower corn. Some of that will be supported by hopefully, an eventual pickup in cattle supplies, but there are other factors. And so we really don't have a level of confidence to tell you, one direction or the other what we think prices will do in 2016 or '17 for that matter.

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